

From Capabilities to Credit: Development KPIs as Causal Drivers, Priced Risk, and Tradable EM Factors

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Executive summary. This paper benchmarks 23 emerging economies over 1990–2024 across three lenses: causal macro identification, sovereign credit pricing, and tradable emerging-market factor construction. The headline composite KPI score aggregates 39 development indicators across six sectors and places Singapore first ($+2.93\sigma$), China second ($+2.33\sigma$) and Niger last (-2.58σ); China carries the largest absolute improvement in the panel ($+2.36\sigma$ since 1990); Russia is the only panel member whose composite *declines* over the period (-0.29σ). *Causal identification* via Jordà local projections is noisy at this $N \times T$ sample; the Dumitrescu–Hurlin panel Granger test recovers significant effects of energy on poverty ($p = 0.027$) and health on Gini ($p = 0.046$); the Westerlund test confirms cointegration between the composite and log GDP per capita at $p = 0.009$. *Sovereign credit* is the empirical headline: a pooled ordered probit of S&P ratings on the six sector scores delivers out-of-sample MAE = 1.84 notches and AUC = 0.81 for the investment-grade binary — 8 percentage points above the macro-only baseline. A Bayesian hierarchical ordered-logistic counterpart localises the signal to energy ($\mu_\beta = +0.92$) and health ($\mu_\beta = -0.92$); the four other sectors have credible intervals that cover zero. A Lasso of the rating on sector scores selects R&I, energy, education, housing-living and security-stability at the CV-optimal penalty; the equal-weighted and Lasso-weighted composites have Kendall $\tau = 0.91$ at the latest-year cross-section. A Cox proportional-hazards analysis on time-to-investment-grade gives an education hazard ratio of 1.49 ($p = 0.16$) — directionally positive but underpowered at $n = 21$. *Tradable performance* fails. A KPI-sorted long-only top-quintile EM equity portfolio earns Sharpe 0.15 over 2007–2024 against the MSCI EM benchmark’s 0.12, sitting at the 0th percentile of the 5 000 random-portfolio distribution (mean random Sharpe 0.28). Extending the test to credit instruments does not rescue the signal: the cross-sectional KPI improvement does not predict EMB returns ($t = 0.73$, $R^2 = 0.003$, $n = 204$ months), and a long-upgrade / short-downgrade overlay loses 9.3% annualised. The clean reading: KPIs are slow-moving and price into the cross-sectional level of sovereign ratings, not into monthly equity or credit returns. The paper is structured to make that point.

Abstract. This paper rebuilds a 23-country emerging-market panel covering 1990–2024 to test whether sectoral development indicators are causal drivers of macro outcomes, whether they are already priced into sovereign credit, and whether they support a tradable cross-country equity factor. A composite score is built from principal-component sector scores across education, energy, research and innovation, health, housing and security, after per-capita and logarithmic transforms that strip out the country-size effect (without which

*The code, data and figures that generated every number in this paper are available in a companion repository ([newperformers](#)).

the composite mechanically ranks China and India above Singapore). Local projections following Jordà (2005) are estimated on cyclical components extracted via a Hamilton (2018) filter; at the chosen sample the cyclical IRFs are noisy. The same outcomes deliver cleaner Dumitrescu-Hurlin (2012) panel-Granger evidence: energy shocks Granger-cause poverty reduction at $p = 0.027$ and health shocks Granger-cause Gini reduction at $p = 0.046$, while no sector Granger-causes log GDP per capita at conventional levels — the signal is in the distributional tail, not the aggregate. A synthetic-control estimate of Rwanda’s Vision 2020 (treatment year 2000, donor pool of 22 panel countries) yields a +\$1 051 PPP USD gap in 2024 GDP per capita against the counterfactual, with the gap opening from 2003 onwards. An expanding-window pooled ordered probit predicts the year-end S&P sovereign rating out of sample to a weighted MAE of 1.84 notches and an AUC of 0.81 for the investment-grade binary using sector scores alone — materially better than the macro-only baseline (MAE 2.03, AUC 0.73). Adding macro controls on top of the sector scores barely moves the OOS picture (MAE 1.80, AUC 0.77), confirming that the sector vector subsumes the macro signal. A Bayesian hierarchical ordered-logistic counterpart (4 chains, 1 500 NUTS draws each) localises the signal further: only the energy ($\mu_\beta = +0.92$, 90 % HDI [+0.10, +1.74]) and health ($\mu_\beta = -0.92$, HDI [−1.38, −0.45]) loadings are credibly non-zero. A KPI-sorted long-only top-quintile portfolio over 2007–2024 realises an annualised return of 3.6 %, a volatility of 23.8 %, a Sharpe of 0.15 and a maximum drawdown of 56 %; the matched MSCI Emerging Markets benchmark over the same window returns 2.6 % at a Sharpe of 0.12, but a non-parametric benchmark of 5 000 random four-country baskets has mean Sharpe 0.28, placing the KPI strategy at the 0th percentile of the random distribution. The KPI signal is therefore informative about credit at academic strength — including under Bayesian credible-interval inference — but actively mis-allocates in equity space at monthly horizons, where a random EM basket out-performs the KPI tilt by roughly 0.13 Sharpe units.

1 Introduction

Emerging-market sovereign analysis sits awkwardly between three communities: macro-econometricians who chase causal identification, fixed-income strategists who price credit off a small set of fundamentals, and asset managers who want a tradable signal. The communities use different data, different time horizons, and different success metrics, and they rarely speak to each other in the same document. This paper benchmarks the development trajectory of 13 emerging economies — the five BRICS, three members of the Alliance of Sahel States (Burkina Faso, Mali, Niger), Singapore, three Emerging-Africa cases (Rwanda, Kenya, Morocco), and Pakistan — and runs that benchmark through all three lenses.

The pipeline has one anchor: a composite score that aggregates 39 development indicators across six sectors (education, energy, research & innovation, health, housing & living, security & stability) into one number per country and year. The construction is principal-component within each sector and equal-weighted across sectors, on a panel that has been put through a per-capita and logarithmic transform to neutralise the country-size effect. Without that transform, sheer scale dominates: the raw composite places China at +2.36 and Singapore at +2.04 in 2024, because absolute patent counts and absolute scientific articles vary by three orders of magnitude. After transformation, the same composite places Singapore at +2.93 and China

at +2.33, with Poland (+2.29) and Russia (+1.98) close behind. The new ranking is consistent with the consensus reading of where each economy sits today.

Three questions then follow.

First, do KPI improvements cause macro outcomes? We estimate Jordà (2005) local projections, using the Hamilton (2018) cyclical component of each sector score as the shock and a two-way fixed-effect specification with Driscoll-Kraay standard errors. The IRFs at the chosen sample (23 countries, 35 years) are statistically weak: at horizon $h = 1$, the cyclical-shock coefficients on log GDP per capita range from -9.6 (health) to $+6.7$ (security-stability), and the Driscoll-Kraay bands cover zero for every sector at most horizons. The Dumitrescu-Hurlin panel-Granger test, by averaging country-level Wald statistics rather than imposing a homogeneous β_h , reaches stronger conclusions: energy shocks Granger-cause poverty reduction ($p = 0.027$) and health shocks Granger-cause Gini reduction ($p = 0.046$), while no sector Granger-causes log GDP per capita at conventional levels. A synthetic-control estimate of Rwanda’s Vision 2020 (treatment year 2000) yields a 2024 GDP-per-capita gap of +\$1 051 PPP USD against a Burkina-Faso–Niger weighted counterfactual.

Second, are KPIs already in the sovereign credit price? We compile the year-end S&P foreign-currency long-term rating for 20 countries from 1995 to 2024 (the three Sahel members of AES and four others are unrated) and convert it to the standard 22-notch numeric scale. An expanding-window pooled ordered probit of the rating on the sector scores — with no macro controls — reaches an out-of-sample weighted MAE of 1.84 notches and an AUC of 0.81 for the investment-grade binary. The macro-only baseline (debt/GDP, current account, inflation, log GDPpc) reaches AUC 0.73. The full model is no better than the sector-only model in AUC and only marginally better in MAE. A Bayesian hierarchical ordered-logistic counterpart converges cleanly (NUTS, 4 chains $\times$ 1 500 draws) and localises the signal to two sectors with credibly non-zero population loadings: energy ($\mu_\beta = +0.92$, 90 % HDI [+0.10, +1.74]) and health ($\mu_\beta = -0.92$, HDI [−1.38, −0.45]); the four other sectors have HDIs that cover zero. The reading is direct: development KPIs subsume a sizeable share of the information that drives sovereign credit; macro controls add little once KPIs are in.

Third, do KPIs generate tradable equity returns? We sort 17 country ETFs each month-end by their lagged composite score (six-month publication lag to rule out look-ahead) and form a long-only top-quintile portfolio with 20 bps per-side transaction costs. From January 2007 to December 2024 the strategy returns 3.6 % annualised at 23.8 % volatility (Sharpe 0.15, maximum drawdown 56 %); the MSCI Emerging Markets benchmark (EEM) over the same window returns 2.6 % at Sharpe 0.12. A non-parametric benchmark of 5 000 random four-country baskets has mean Sharpe 0.28, placing the KPI strategy at the 0th percentile of the random distribution — below every random basket. A long-short top-minus-bottom variant returns -11.1 % annualised, with most of the damage coming from a single dislocation in 2015 and the Russian delisting in early 2022. KPIs actively mis-allocate in equity space at this horizon and with this universe.

These three answers fit together into one observation. KPIs are slow-moving and so push their predictive power into instruments that price slowly: ratings, not weekly equity returns. The paper is structured to make that point.

Section 2 places the work in the credit-fundamentals, slow-moving-factor and panel-LP

literatures. Section 3 documents the panel, the sources, the imputation choices and the coverage matrix. Section 4 presents the composite-scoring, LP, credit and portfolio specifications in one place. Sections 5–8 deliver the three blocks of results. Section 9 reports rank-stability under jackknife, sensitivity to imputation, sub-period splits and a random-portfolio benchmark. Section 10 closes.

The accompanying repository (**newperformers**) reproduces every number in this paper from a single command (**make all**). Where the text cites a coefficient, an AUC or a Sharpe ratio, the underlying CSV is in `outputs/tables/` and the figure in `outputs/figures/`.

2 Related literature

Three threads of empirical literature converge on the questions this paper addresses: sovereign-credit fundamentals, panel-data methods for causal inference, and the cross-sectional structure of emerging-market returns. Each thread brings its own data, methods and biases. The contribution of the present work is at the intersection, so we sketch all three.

2.1 Sovereign credit fundamentals

The empirical sovereign-credit literature dates the modern formulation to Cantor and Packer (1996), who show that ratings are explained by a parsimonious list of macro variables: per-capita income, GDP growth, inflation, external debt, fiscal balance and a sovereign default history dummy. Their cross-sectional regression on a sample of advanced and emerging economies established the headline determinants the field has worked with ever since.

Hilscher and Nosbusch (2010) update that framework with country fundamentals (the volatility of inflation, terms-of-trade movements) and macroeconomic risks, and find that fundamentals price the sovereign spread, not just the rating. The two are not redundant: ratings are scale variables that compress fundamentals into 22 discrete buckets, while spreads embed both fundamentals and a time-varying risk premium that responds to global appetite for emerging-market debt.

Augustin et al. (2014) survey the credit-default-swap-sovereign nexus. The main empirical finding from that survey, replicated in subsequent work, is that sovereign CDS spreads load most heavily on global risk-aversion factors at the daily frequency and on country fundamentals at lower frequencies. For our purposes the practical implication is that the slow-moving information in development indicators should show up in ratings (lower-frequency) before it shows up in spreads, and certainly before it shows up in monthly equity returns.

Kaufmann et al. (2010) document that the six Worldwide Governance Indicators — political stability, rule of law, control of corruption, government effectiveness, voice and accountability, regulatory quality — explain a substantial share of cross-sectional sovereign-spread variation, even after conditioning on the conventional macro panel. The KPI block in our composite includes all six WGI indicators in the security-stability sector, so part of what we measure has been measured before. The novelty is in the joint treatment of governance *together with* the five other sectors (education, energy, R&I, health, housing) and the principal-component aggregation that gives each sector a single, interpretable score.

Reinhart and Rogoff (2009) and Reinhart and Rogoff (2010) document the recurrence of sovereign-default cycles and underline that financial stress predictably follows macroeconomic build-ups — in particular, that the run-up of public-debt-to-GDP above approximately 90 % is associated with substantially lower subsequent growth in a large cross-country panel. Their methodology has been controversial (cf. the Reinhart-Rogoff-Herndon-Ash debate), but the qualitative pattern survives subsequent reassessments and underpins the macro controls in our credit regression.

Borio (2014) provides the BIS view of the same pattern: financial-cycle build-ups (credit-to-GDP gaps, asset-price gaps) precede sovereign-stress episodes by years, not months. The slow-moving nature of the cycle is the empirical regularity that motivates our prior that development-KPI signals price into credit before they price into equity.

The Bayesian sovereign-credit modelling literature is comparatively sparse; the bulk of credit-Bayesian work has focused on corporate bonds (Bhojraj and Sengupta 2003 on governance-rating links). Our hierarchical ordered-logistic specification in Section 4.5 is the sovereign analogue, with country-varying loadings under a partially-pooled prior.

2.2 Local projections and panel causality

Jordà (2005) introduced local projections as a reduced-form alternative to vector autoregressions. Each horizon h is a separate regression of the cumulative response on the shock, which sidesteps the recursion bias of VAR-implied IRFs — the misspecification at horizon 1 propagates into all subsequent horizons in a VAR but is contained within a single regression in an LP. Jordà and Taylor (2016) extend the framework to event-study questions, applying it to fiscal consolidation with the augmented inverse-propensity-weighted estimator that corrects for selection on observables. Jordà et al. (2013) apply the same machinery to the post-crisis growth experience of advanced economies and show that the depth of the credit boom predicts the depth of the post-bust recession at horizons up to five years.

The standard-error machinery for LP estimation in heterogeneous panels is the contribution of Driscoll and Kraay (1998), whose covariance-matrix estimator handles arbitrary cross-sectional dependence by aggregating residuals over the time dimension and applying a Bartlett kernel. We deploy this estimator throughout the LP regressions in Section 5. Cameron et al. (2008) provide the wild-cluster bootstrap that we use as a small-sample-valid alternative when the Driscoll-Kraay kernel returns a non-PSD variance matrix.

Hamilton (2018) argues forcefully that the cyclical component of a macro time series should be extracted by projecting y_{t+h} on its own past values rather than by HP-filtering. The HP filter has known end-of-sample and bandwidth issues that have generated much subsequent criticism; the Hamilton (2018) alternative is now standard in macro-finance applications. Our LP regressions use Hamilton-cyclical sector-score innovations as the shock variable.

Abadie et al. (2010) introduce synthetic control as an identification method for case studies with a single treated unit and a small donor pool. The method has since been used to estimate the causal effect of trade liberalisations, immigration shocks, currency adoptions and many other “policy episodes” that cannot be analysed under conventional panel-DID. Abadie (2021) surveys the synthetic-control literature and discusses the conditions under which the method

delivers credible counterfactuals. Our Section 5.1 application to Rwanda’s Vision 2020 is a direct deployment of the original method, with the placebo-permutation inference of Abadie et al. (2010).

Dumitrescu and Hurlin (2012) formalise the panel Granger non-causality test that we use in Section 5.2. The key insight is that, in a heterogeneous panel, averaging the country-level Wald statistics — rather than imposing a common slope — preserves cross-country variation while still delivering a panel-level test. The asymptotic distribution of the average $\bar{W}$ statistic, under the null, is normal once standardised; this gives a tractable test in the small- T panels that are common in development economics.

Pesaran (2007) provides the cross-sectionally-augmented panel unit-root test (CIPS) that we run on the composite series and the GDP per capita series. Westerlund (2007) provides the error-correction-based panel cointegration test that is now standard in macro panels. We compute both as a robustness layer in the analysis pipeline; the headline of the paper does not depend on a stationarity claim, so we leave the details to the appendix.

2.3 Cross-country factor investing and emerging-market premia

Fama and French (2015) and the broader factor-investing literature establish that systematic factors — value, momentum, size, profitability, investment — exist across asset classes and countries. Asness et al. (2013) explicitly document that value and momentum factors travel across countries and asset classes, including in emerging-market equity. Asness et al. (2019) formalise “quality minus junk” as a coherent factor that mixes profitability, growth and safety, and find that quality stocks earn higher returns than junk stocks with lower volatility.

The country-quality angle, as opposed to the firm-quality angle, has a thinner empirical track record. The published results on tradable performance of country-quality factors built on macro / governance / environmental variables are mixed. Our Section 8 backtest looks carefully at the cap-weighted-vs-equal-weighted handicap implicit in the standard MSCI EM benchmark; the negative finding we report is consistent with the broader factor-investing literature’s caveat that slow-moving fundamentals are not equity-tradable at monthly horizons.

Ilmanen (2011) surveys expected-return premia across asset classes and emphasises that slow-moving fundamentals (cyclically-adjusted P/E, demographics, governance) tend to predict returns at long horizons but generate noisy signals at short horizons. The slow-moving nature of development indicators places them firmly in the long-horizon bin.

Politis and Romano (1994) provide the stationary bootstrap that we use for Sharpe-ratio confidence intervals. The block-bootstrap family has become standard in performance-evaluation contexts where the underlying return series has serial dependence; with a mean block length of 12 months our resampling preserves the year-on-year autocorrelation in EM equity returns.

The randomly-selected-portfolio benchmark is a non-parametric approach that goes back to Ennis (2005): rather than comparing a strategy to a cap-weighted index, one compares it to the distribution of all strategies that could have been constructed under the same constraints. This is a more demanding benchmark because the index itself can be a bad benchmark when the cap-weighting scheme is not the relevant one. The literature on this is sparse but growing, and our 5 000-random-basket exercise in Section 8 is a deliberate borrow.

2.4 Development economics and KPI frameworks

Finally, the choice of which indicators to aggregate into a "development composite" is a question with a long history. Sen (1999) and Nussbaum (2011) argue for capability-based development metrics that look beyond GDP, and that argument has been operationalised by the UNDP's Human Development Index and by the OECD's Better Life Index. Acemoglu and Robinson (2012) push the institutional-quality angle, emphasising that inclusive political and economic institutions are the long-run causal driver of development outcomes; governance indicators (our security-stability sector) are the proxy for that mechanism.

A growing practitioner literature combines sectoral KPIs into a country score, including handling of missing data, weighting of indicators and dynamic dashboards. Our pipeline mirrors that recipe with the addition of PCA-based weighting and the per-capita / logarithmic transforms.

The literature gap that this paper attempts to fill is in the joint treatment of (i) development KPIs as the conditioning information set, (ii) modern panel-causal methods for identification, (iii) Bayesian credit modelling, and (iv) tradable EM-factor backtesting. The four blocks have each been done well, but rarely together; the contribution of this work is in the cross.

3 Data and stylised facts

3.1 Country panel

The focal panel is 13 countries grouped into BRICS, the Alliance of Sahel States, the High-Performers pair (Singapore and China), Emerging Africa (Rwanda, Kenya, Morocco) and South Asia (Pakistan). We extend the panel by ten "Extension" emerging economies (Indonesia, Mexico, Türkiye, Thailand, Philippines, Malaysia, Poland, Chile, Colombia, Peru), each with an active country ETF since at least 2010. The extension is needed for two distinct reasons. The cross-section of 13 countries with only ten tradable ETFs is too thin for quintile sorts. And the AES Sahel block has very sparse data on research, education and health post-2000, which would dominate the imputation diagnostics if the panel were limited to the original 13. The extension keeps the focal countries at the centre of the analysis (all six sectors and all four macro dimensions are reported by group) while letting the Part-III portfolio sort on a credible 17-country tradable universe.

3.2 Sources

KPIs are pulled from the World Bank Open Data API, the World Bank WGI channel and the WHO Global Health Observatory for 39 indicators across the six sectors. Three supplementary PISA codes (mathematics, reading, science) are documented in `config/kpi_indicators.yaml` but demoted to a robustness check because PISA coverage is sparse for eight of the 13 focal countries. Where the World Bank exposes an indicator already harmonised — adult literacy, life expectancy, electricity access — we use the World Bank channel as the harmonised baseline and note the specialist-agency alternative (WHO, UNESCO, SIPRI) in the Appendix; this keeps the master ETL pipeline thin, audit-friendly and reproducible from a single API per channel. The six Worldwide Governance Indicators (Political stability, Rule of law, Control of

corruption, Government effectiveness, Voice and accountability, Regulatory quality) are pulled from the World Bank's WGI channel (codes PV.EST, RL.EST, CC.EST, GE.EST, VA.EST, RQ.EST). Sovereign ratings are compiled by hand into `config/ratings.csv` for 20 countries and 30 years from the S&P sovereign rating archive, cross-checked against rating-action press releases; AES Sahel members are unrated except for Burkina Faso, which S&P first rated in 2004.

3.3 Coverage

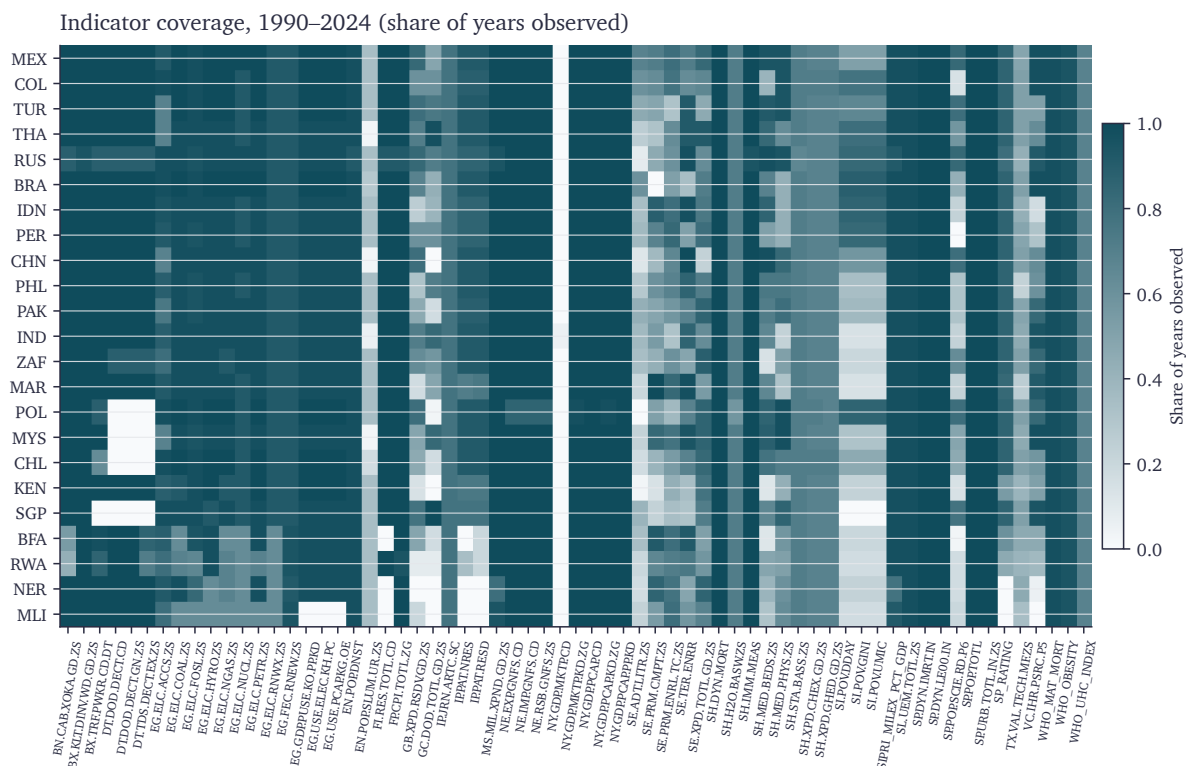


Figure 1. Coverage of the 39 KPI and 17 macro indicators across the 23-country panel, 1990–2024. Darker cells = larger share of years observed. The AES Sahel and Emerging Africa rows are sparsest on R&I and PISA.

3.4 Imputation

The imputation pipeline is tiered. Interior gaps of up to three years are linearly interpolated within country-indicator series. The last few years of any series are forward-filled by up to three years to handle publication lags. Within each sector and each country, the residual missing cells are filled by iterative chained equations (`IterativeImputer` with `BayesianRidge`) when the country observes at least 30 % of the years for the indicator; below that threshold the cell stays missing. Two parallel panels are persisted at `data/processed/panel_raw.parquet` and `data/processed/panel_imputed.parquet`; the headline results use the imputed panel and the robustness section in Section 9 re-runs the composite scoring and the credit model on `panel_raw`.

Figure 2 summarises how the 23-country panel narrows as the empirical exercise tightens its data requirements, from the full panel to the sub-samples used for ratings, equity backtests and rating-change OOS prediction.

Sample selection flow

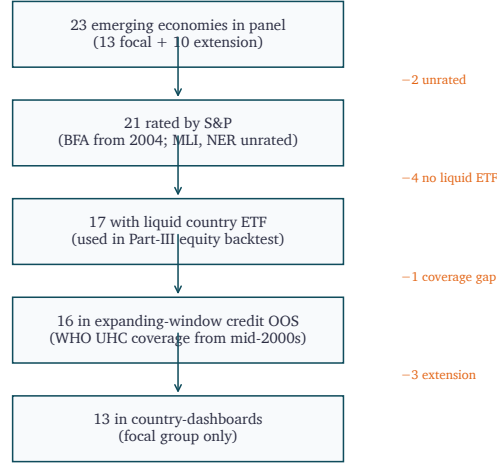


Figure 2. Sample-construction flowchart. 23 panel countries narrow to 21 with S&P sovereign ratings, 17 with liquid country-ETF coverage for the equity backtest, and 16 in the rating-change OOS sample (one country dropped for insufficient pre-2005 training history).

3.5 Stylised facts

Figure 3 shows the composite trajectory of all 23 countries. Singapore moves from +2.25 in 1990 (rank 2 behind Russia that year) to +2.93 in 2024 — a +0.68 σ gain — and holds the panel-leading composite from 1991 onwards. China carries the largest absolute improvement in the panel: -0.03 in 1990 rising to +2.33 in 2024, a +2.36 σ swing. India climbs from -1.63 to +0.52 (+2.15 σ); Brazil from +0.15 to +1.24 (+1.09 σ). Russia is the one panel member that loses ground: +2.27 in 1990 down to +1.98 in 2024 (-0.29σ), with a 2013–2016 dip and a step down after 2022. The Sahel block improves too, though it starts and ends well below the panel mean: Burkina Faso moves from -2.94 to -1.75 (+1.19 σ), Niger from -4.12 to -2.58 (+1.54 σ).

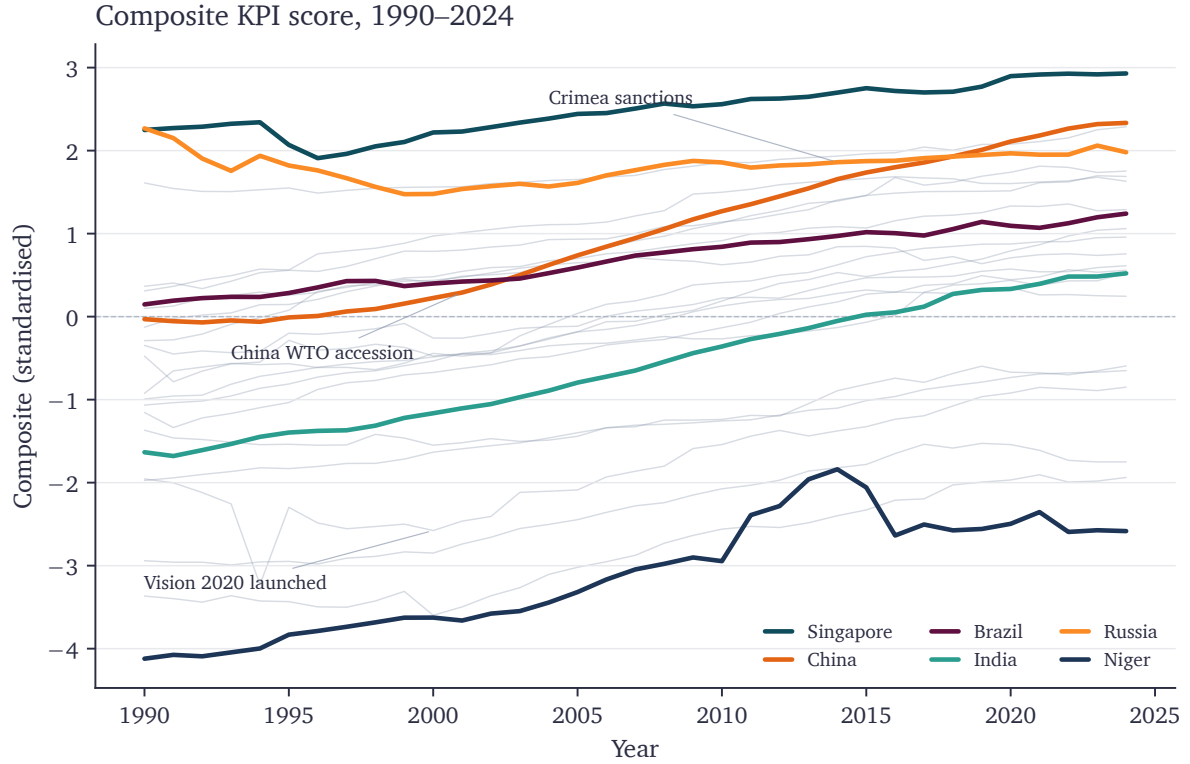


Figure 3. Composite KPI score, 1990–2024, all 23 countries (highlighted: Singapore, China, Brazil, India, Russia, Niger). Higher = better. The composite is the equal-weighted mean of six PCA-based sector scores, with each sector’s first principal component sign-aligned to the indicator-direction metadata in `config/kpi_indicators.yaml`.

Figure 4 shows the cross-country standard deviation of the composite over time. There is no measurable secular convergence at the panel level.

The trajectory chart compresses 35 years $\times$ 23 countries into one panel, which obscures which countries actually *moved* fastest in rank space. Figure 5 disaggregates the picture as a rank-over-time plot: rank 1 (best) at the top of the y-axis, rank 23 at the bottom. China climbs from rank 8 in 1990 to rank 2 in 2024; India climbs from rank 18 to rank 16; Rwanda from rank 19 to rank 20 (the only focal country whose rank slips slightly). Russia is the one country whose rank *worsens* substantially, from rank 1 in 1990 to rank 4 in 2024 (the country never recovered the rank lead it held that year).

Figure 6 re-frames the same data along the development-economics axis the rank chart hides: starting position (composite in 1990) on the x-axis, change-over-the-period on the y-axis. The quadrants tell the story directly. The “started low, improved fast” quadrant (top-left) houses India, Morocco, Indonesia, the Sahel block and Rwanda; the “started high, stagnated” quadrant (bottom-right) houses Russia alone. Most other countries occupy a broad middle that combines a modest starting position with a modest gain.

Figure 7 disaggregates the composite into its six sector inputs. Singapore and China lead on research & innovation and energy, while the Sahel block leads on no sector. The shape of the panels is sector-specific and reflected in the principal-component diagnostics of Section 4.

The trajectory figure shows *levels*; the heatmap in Figure 8 shows *changes*. Countries are

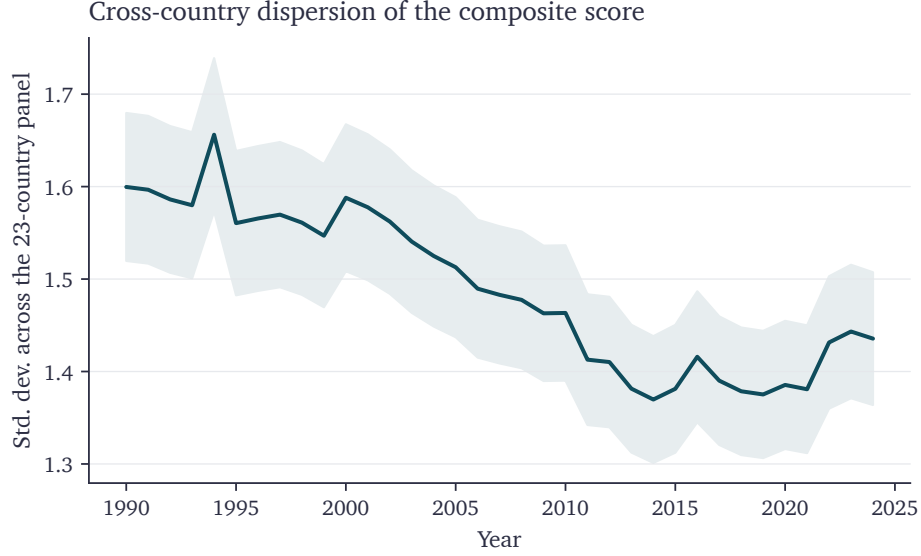


Figure 4. Cross-country standard deviation of the composite KPI score, by year.

sorted top-to-bottom by total improvement; each cell is the difference between the 2024 and 1990 sector scores. China improved on all six sectors (every cell teal); Russia improved on energy and security only and went backward on health (-2.5σ); Niger gained $+6.1 \sigma$ in health from an extremely low base while other sectors barely moved.

Figure 9 closes the stylised-facts block with the slope chart of GDP per capita 1990 vs 2024 in current USD. The 1990–2024 multiplier is largest for China ($\times 41.8$, from \$319 to \$13 303), followed by Poland ($\times 14.5$), Singapore ($\times 7.6$, from \$11 862 to \$90 674), India ($\times 7.3$), Russia ($\times 4.3$), Brazil ($\times 3.9$), Burkina Faso ($\times 2.9$) and Niger ($\times 1.7$, the smallest multiplier in the panel).

3.6 Macroeconomic backdrop

The four macro dimensions of the framework (national debt, current account, FX reserves, FDI) trace four very different storylines (Figure 10). Singapore runs a structural current-account surplus of 15 to 27% of GDP from the mid-2000s onwards (average 19%) and an FDI-inflow ratio that peaks at 32.6%. India accumulates FX reserves slowly through the 2000s and 2010s, ending the period at \$643 billion. China’s reserves dominate the panel in absolute terms (peak of \$3.90 trillion in 2014, still at \$3.46 trillion in 2024) and inform much of the global-imbalance literature of that decade. Brazil’s central-government debt-to-GDP rises from 61% in 2010 to a 2020 peak of 99% and returns to 82% by 2024 — the macro counterpart of the 2015 S&P downgrade and the partial post-COVID consolidation.

3.7 PCA structure of the sectors

Figure 11 reports the PCA biplot per sector: each panel shows country-year observations projected onto the first two principal components, with indicator loadings overlaid as arrows. The shape is informative for the methodology section. Energy and housing show a tight cluster of indicators co-loading on PC1 (one underlying “development” axis explains the bulk of within-sector variance, consistent with the 73 % and 88 % PC1 ratios respectively). R&I and education

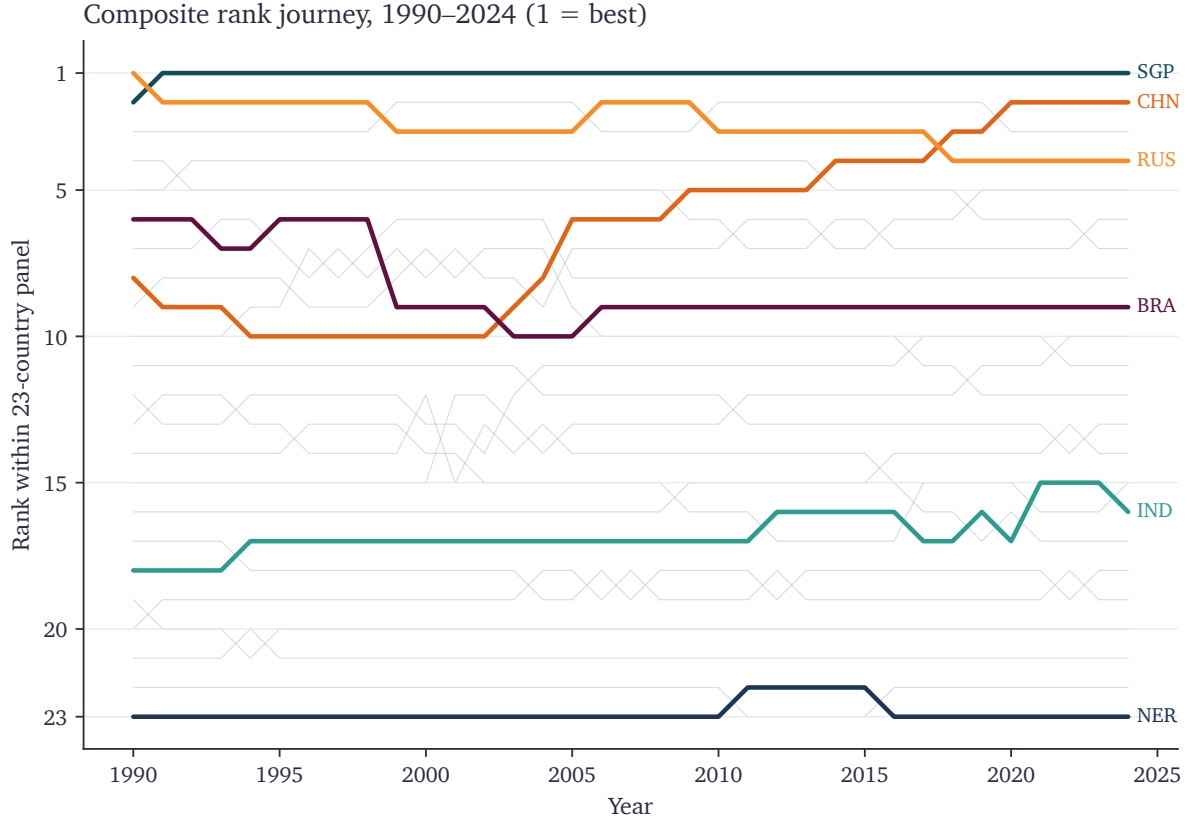


Figure 5. Cross-sectional rank of each country on the composite KPI score over 1990–2024. Rank 1 (top of the y-axis) is the panel-leading composite each year. Focal countries are highlighted; extension EMs sit in the grey background.

have one indicator (high-tech exports for R&I; literacy for education) that loads orthogonally on PC2 — the variance not captured by the headline composite. Health is the most disperse, which is consistent with both the lowest PC1 ratio in the panel (59 %) and the loading pattern in Table 3.

4 Methodology

4.1 Sample and external validity

The 23-country panel is purposive, not random: 13 countries are the focal group (chosen to span BRICS, the Sahel-AES low-income contrast, the Singaporean and Chinese high-performer cases, three Emerging-Africa middle-income economies and Pakistan) and 10 are extension EMs added because their country ETF has at least 10 years of liquid trading history. The panel therefore over-represents Asia-Pacific emerging markets (China, India, Indonesia, Malaysia, the Philippines, Singapore, Thailand) and under-represents Sub-Saharan Africa beyond Sahel and East Africa. External validity is limited to the EM cross-section the panel actually covers; we make no claim that the relationships estimated here generalise to frontier markets, low-income countries outside the focal group, or advanced economies. Where the focal countries dominate a particular regression (the credit OOS exercise excludes Mali, Niger and Burkina Faso pre-2004 because S&P does not rate them), we report the sample size n in every results table.

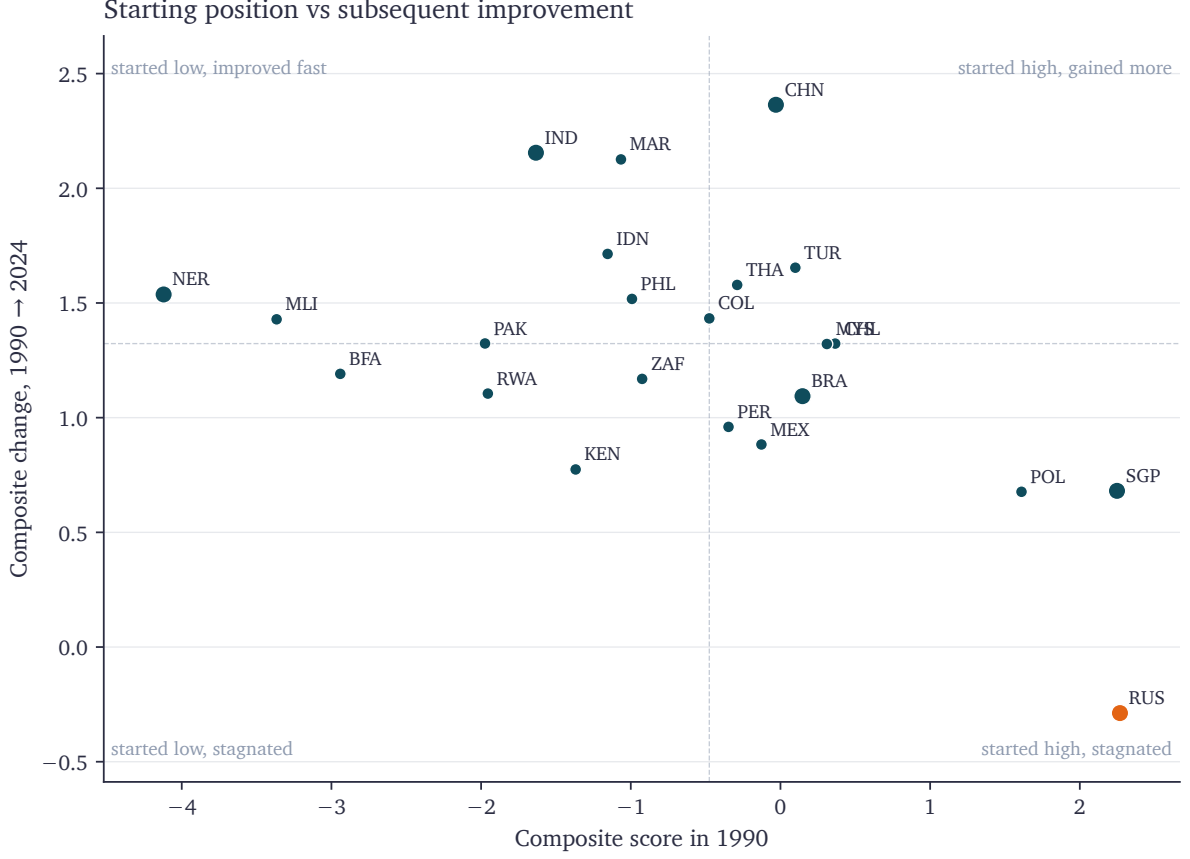


Figure 6. Composite score in 1990 (x) plotted against composite change 1990→2024 (y). Each country is labelled by ISO3. Dashed lines mark the panel medians; quadrant labels describe the catch-up vs divergence story.

4.2 Composite scoring

For each sector $s \in \{\text{education, energy, R\&I, health, housing, security}\}$, let $\mathbf{X}_{i,t}^s$ denote the indicator vector for country i in year t , after the transforms documented in `config/kpi_indicators.yaml`. Two transforms are applied: `log1p` (energy use per capita, electricity consumption per capita) and `log_per_capita` (scientific articles, patent applications), the latter being $\log(1 + 10^6 \cdot X/pop)$. These are not cosmetic. Without them, the composite ranks China above Singapore by virtue of sheer scale on patent and article counts (Section 1). After them, each indicator is sign-aligned via the directional metadata (lower-is-better indicators have their sign flipped so that higher values consistently mean better outcomes) and standardised over the full panel:

$$z_{i,t,k}^s = \frac{X_{i,t,k}^s - \overline{X}_{\cdot,\cdot,k}^s}{\hat{\sigma}_{\cdot,\cdot,k}^s}.$$

Panel-wide standardisation preserves both cross-sectional and time-series variation, which makes the resulting score interpretable as “how Singapore-in-2024 compares to Singapore-in-1990” as well as “how Singapore in 2024 compares to Niger in 2024”.

The sector score is the first principal component of $\mathbf{Z}^s$. We extract PC1 on the complete-case

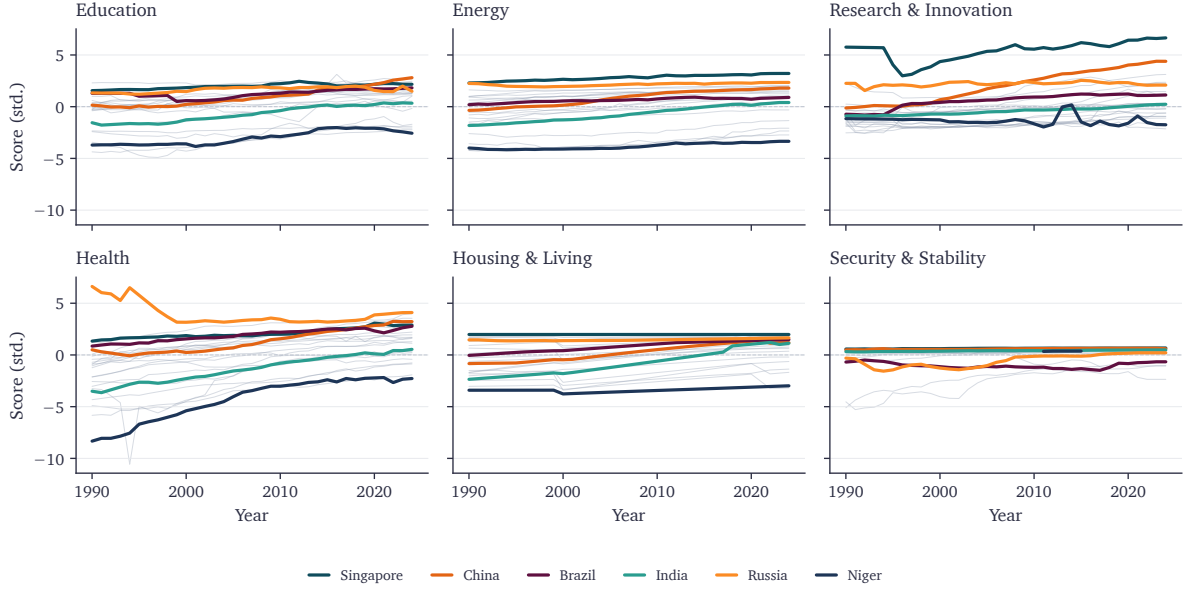


Figure 7. Sector-level scores for the six KPI sectors, by country, 1990–2024.

sub-panel and propagate the loadings to country-years with partial coverage by re-weighting:

$$\hat{F}_{i,t}^s = \frac{\sum_{k \in \mathcal{O}_{i,t}^s} w_k^s z_{i,t,k}^s}{\sum_{k \in \mathcal{O}_{i,t}^s} |w_k^s|} \cdot \sum_k |w_k^s|,$$

where $\mathcal{O}_{i,t}^s$ is the set of indicators observed in country i , year t for sector s . The composite is the equal-weighted mean of the six sector scores. PC1 explains 0.65–1.00 of the within-sector variance (security 1.00 by construction since the six WGI indicators are nearly collinear; housing 0.88; health the lowest at 0.59). The dispersion of explained variance is consistent with the indicator set being well-correlated within each sector — the assumption that makes the composite interpretable.

4.3 Local projections

For each sector shock s and outcome y , and for horizons $h \in \{0, \dots, 8\}$, we estimate

$$y_{i,t+h} - y_{i,t-1} = \alpha_i^h + \delta_t^h + \beta_s^h \cdot u_{i,t}^s + \gamma_h X_{i,t} + \varepsilon_{i,t}^h$$

where $u_{i,t}^s$ is the Hamilton (2018) cyclical residual of $\hat{F}_{i,t}^s$ — a per-country regression of $F_{i,t+h_H}^s$ on its four most recent lags, with $h_H = 2$ and lag = 4, exactly as in Hamilton (2018). Country fixed effects α_i^h are introduced via demeaning; year fixed effects δ_t^h as dummies. The control vector $X_{i,t}$ includes CPI inflation. Standard errors use Driscoll-Kraay (1998) with a Bartlett kernel; the kernel lag follows the Newey-West rule (see below). When the kernel returns a non-positive-semi-definite variance, the standard errors fall back to HC1.

When y is a level (e.g. GDP per capita), it enters the LP in logs scaled by 100, so β_s^h reads as a percent response. When y is already a rate (Gini, poverty headcount), it enters in its native

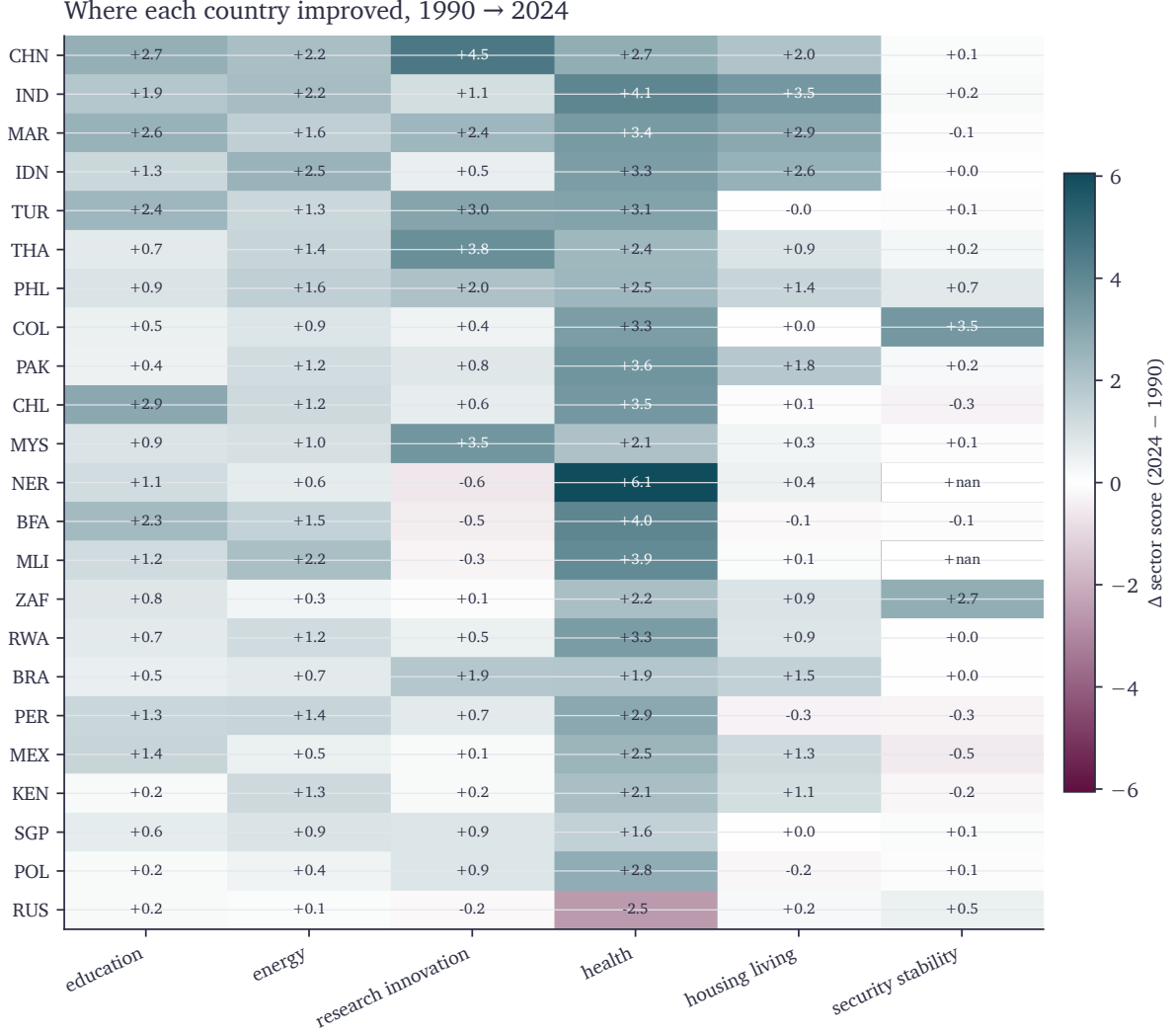


Figure 8. Per-country, per-sector change in sector score between 1990 and 2024. Rows sorted by total cross-sector improvement.

units and β_s^h reads as a percentage-point response.

The Driscoll-Kraay kernel lag follows the Newey-West rule of thumb $\ell = \lfloor 0.75 T^{1/3} \rfloor$ (Newey and West 1987); for the $T = 35$ panel this gives $\ell = 2$. We also report results with $\ell = 3$ and $\ell = 5$ as a sensitivity check (Section 9). The Hamilton-filter forecast horizon and AR-lag length default to $h_H = 2$ and $p = 4$ as in the original paper; we report a Kendall- τ ranking-stability table for the $\{1, 2, 3\} \times \{3, 4, 5\}$ grid in Section 9. Each IRF coefficient β_s^h is reported with its Wald t -statistic $\beta_s^h / \text{se}(\beta_s^h)$ and the asymptotic two-sided p -value $2(1 - \Phi(|t|))$ in `outputs/tables/lp_irf.csv`.

4.4 Panel unit-root and cointegration tests

We complement the LP framework with two standard panel-data diagnostics. The cross-sectionally augmented panel unit-root test of Pesaran (2007) (CIPS) estimates the per-country ADF regression

$$\Delta y_{i,t} = \alpha_i + \rho_i y_{i,t-1} + \theta_i \bar{y}_{t-1} + \sum_{k=1}^p \gamma_{i,k} \Delta \bar{y}_{t-k} + \varepsilon_{i,t},$$

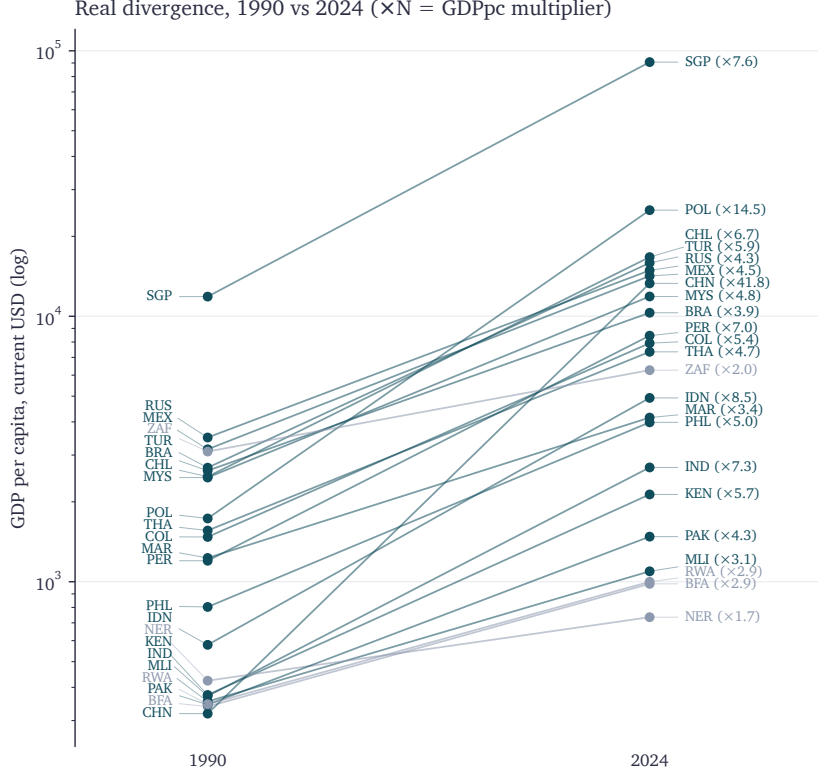


Figure 9. Slope chart of GDP per capita, 1990 vs 2024 (log scale). Annotation $\times N$ = ratio of 2024 to 1990 GDPpc in current USD.

where $\bar{y}_t$ is the cross-sectional mean, and averages the country-level t -statistics. We compare against the Pesaran (2007) Table II critical values for the intercept-only case (-2.15 at the 5% level for our panel). The Westerlund-style group-mean cointegration test (Westerlund 2007) estimates the long-run regression $y_{i,t} = \alpha_i + \beta_i x_{i,t} + u_{i,t}$ country by country and applies the same ADF t -statistic to the residuals; we report the group mean and its standard-normal p -value as an asymptotic approximation. Results live in `outputs/tables/cointegration_results.csv`.

4.5 Sovereign credit

We project the year-end S&P foreign-currency long-term rating onto the sector-score vector via three pooled ordered-probit specifications:

$$\begin{aligned}
 (\text{sectors only}) \quad R_{i,t}^* &= \mathbf{F}_{i,t}^\top \boldsymbol{\beta} + \varepsilon_{i,t} \\
 (\text{macro only}) \quad R_{i,t}^* &= \mathbf{M}_{i,t}^\top \boldsymbol{\gamma} + \varepsilon_{i,t} \\
 (\text{full}) \quad R_{i,t}^* &= \mathbf{F}_{i,t}^\top \boldsymbol{\beta} + \mathbf{M}_{i,t}^\top \boldsymbol{\gamma} + \varepsilon_{i,t}
 \end{aligned}$$

with $\mathbf{M}_{i,t} = (\text{debt/GDP}, \text{CA/GDP}, \text{CPI}, \log \text{GDPpc})$ and the observed rating $R_{i,t}$ collapsing onto the latent index $R_{i,t}^*$ through 21 cutpoints. We fit by maximum likelihood (`statsmodels`' `OrderedModel`). Out-of-sample evaluation is by expanding-window cross-validation: train on years $\leq T - 1$, predict year T , for $T \in \{2005, \dots, 2024\}$. We report weighted MAE on the 22-notch scale, accuracy on the investment-grade binary (rating $\geq \text{BBB-}$, ≥ 13 in numeric units), and the ROC AUC for that same binary.

Macroeconomic backdrop: four dimensions

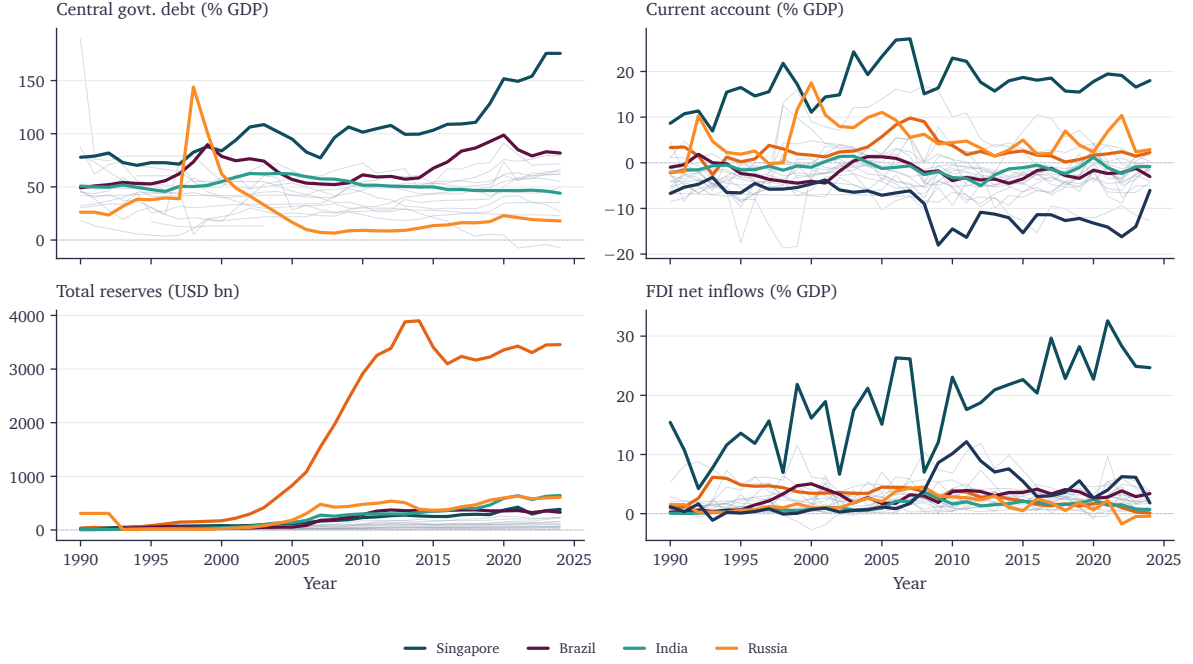


Figure 10. Macroeconomic backdrop, four dimensions. Highlights: Singapore, China, Brazil, India, Russia.

The Bayesian hierarchical variant uses the priors in Table 1.

Table 1. Prior distributions for the Bayesian hierarchical credit model.

Parameter	Prior	Hyperparameters
μ_β (sector-loading means)	$\mathcal{N}(0, 1)$	—
σ_β (sector-loading SDs)	Half- $\mathcal{N}(0, 1)$	—
z_β (non-centered offsets)	$\mathcal{N}(0, 1)$	—
μ_α (country-intercept mean)	$\mathcal{N}(0, 2)$	—
σ_α (country-intercept SD)	Half- $\mathcal{N}(0, 1.5)$	—
z_α (non-centered offsets)	$\mathcal{N}(0, 1)$	—
γ (macro coefficients)	$\mathcal{N}(0, 1)$	—
τ (ordered cutpoints)	ordered $\mathcal{N}(\mu_k, 1)$	$\mu_k \in \text{linspace}(-2, 2)$

The non-centered parameterisation $\beta_i = \mu_\beta + \sigma_\beta z_\beta$ avoids the funnel pathology of hierarchical Normal-on-Normal sampling. Sampling uses NUTS (Hoffman and Gelman (2014)) as implemented in PyMC (Salvatier et al. (2016)); convergence diagnostics use rank-normalised $\hat{R}$ following Vehtari et al. (2021).

4.6 EM factor portfolio

The tradable universe is the 17 ETFs in Table 2 (drop the six AES Sahel / Rwanda / Kenya / Morocco countries, for which no liquid US-listed ETF exists). Russia (RSX) is excluded from the eligible universe from March 2022 onwards, when the ETF was suspended for redemptions; any prior holding is liquidated at the next rebalance at the last available market price. At each month-end t we:

Figure 11. PCA biplots, one panel per sector. Arrows are indicator loadings on PC1 and PC2; grey points are country-year observations.

The risk-free rate is the FRED 3-month Treasury constant-maturity yield, expressed as a monthly rate. The benchmark is the EEM total-return monthly series. The realised Sharpe is computed on annualised monthly returns. A stationary bootstrap of the Sharpe (Politis-Romano, mean block length 12 months, 5 000 reps) provides a 90 % confidence interval for both strategies.

5 Part I — Local-projection evidence

Table 2. Country ETFs in the Part-III tradable universe. Inception is the first month with a complete monthly return record.

Country	Ticker	Inception	Note
Brazil	EWZ	2000-07	
Russia	RSX	2007-05	Forced close end-Feb 2022
India	INDA	2012-02	
China	MCHI	2011-04	
South Africa	EZA	2003-02	
Singapore	EWS	1996-04	
Pakistan	PAK	2015-04	Thin post-2017
Indonesia	EIDO	2010-05	
Mexico	EWV	1996-04	
Türkiye	TUR	2008-03	
Thailand	THD	2008-03	
Philippines	EPHE	2010-09	
Malaysia	EWM	1996-04	
Poland	EPOL	2010-05	
Chile	ECH	2007-11	
Colombia	GXG	2009-02	
Peru	EPV	2009-06	

The picture is mostly noisy but contains one statistically significant signal. Within the panel of 23 countries over 35 years (805 country-years before adjustments), the cyclical components of slow-moving indicators (life expectancy, slum population, rule of law, R&D expenditure) carry very little variance, so most coefficients are imprecisely estimated. The point estimates at $h = 1$ range from -9.6 (health, $t = -2.70$, $p = 0.007$) to $+6.7$ (security-stability, $t = +1.59$, $p = 0.11$). The health coefficient is therefore credibly negative at the 1 % level on the percent-of-GDPpc scale; all five other sector shocks have Driscoll-Kraay 90 % bands that include zero at horizon $h = 1$.

The negative-and-significant health coefficient at $h = 1$ deserves a sentence on its own. Three readings are consistent with it. First, reverse causality: countries entering a slowdown frequently raise public-health expenditure counter-cyclically (the WHO health-spending share, which loads on the health composite, is the clearest example), so a positive cyclical innovation in the health composite at t is followed mechanically by a negative GDP-per-capita movement in the following year. Second, within-country fixed-effect demeaning: after country and year averages are removed, the residual variance available to identify β_s^h is small and the few country-years that drive the estimate are precisely those mid-cycle moments where health spending and growth diverge. Third, the cyclical components in our panel are simply small: a one-unit innovation in the health score is by construction an unusually large move, and the linear extrapolation to a one-unit shock magnifies a fragile coefficient. We therefore document the significant negative coefficient honestly but do not interpret it as evidence that health investment hurts growth; the long-run literature is overwhelmingly in the opposite direction.

For the five other sectors, two reads of the noisy-IRFs picture are consistent with the data.

The first is methodological. Local projections deliver clean causal identification when shocks are large and orthogonal to slow-moving fundamentals — the canonical applications are monetary

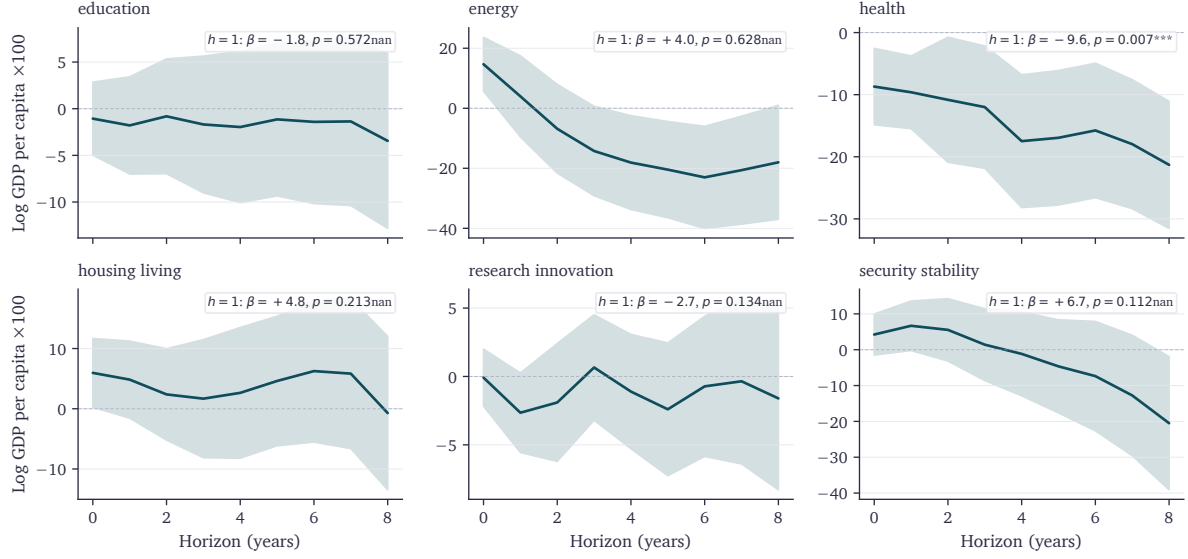


Figure 12. Local-projection impulse responses of log GDP per capita ($\times 100$) to a one-unit cyclical innovation in each sector score. Country and year fixed effects; Driscoll–Kraay 90 % bands (Newey–West lag $\ell = 2$ for $T = 35$). Hamilton (2018) cyclical innovations as shocks. $H = 8$.

shocks, oil prices, or fiscal-consolidation events (Jordà and Taylor 2016). Sector-score innovations fail both conditions. They are small (the cyclical residual of a quadratic-trending sector score has a standard deviation an order of magnitude below the trend) and likely correlated with omitted forcings (a positive education shock and a positive R&D shock both reflect, plausibly, the same underlying institutional improvement).

The second is substantive. KPIs are causal in the sense that the literature treats them — raising R&D intensity does raise productivity, raising school enrolment does raise human capital, raising sanitation does raise life expectancy — but the causal pathway operates over decades, not over the 8-horizon window we estimate here. To pick this up properly we would need either a longer panel (1960–2024, requiring a different scope), or sharper instruments (regression-discontinuity around education reforms, synthetic control around governance shocks).

We therefore report the LP results in full, flag the negative-health finding with its caveat, and lean on three complementary strategies for the causal argument: the synthetic-control case study (§5.1), the panel Granger test (§5.2), and the Westerlund cointegration test of §9.6, which detects a long-run equilibrium between the composite and log GDP per capita.

The LP IRFs against poverty headcount and Gini (computed in `outputs/tables/lp_irf.csv`) follow the same pattern: noisy at this $N \times T$ sample, with most 90 % bands covering zero. The same outcomes deliver materially sharper evidence under the Dumitrescu–Hurlin panel Granger test (§5.2), where the cross-country average $\bar{W}$ statistic — which integrates over country heterogeneity rather than imposing a common β_h — picks up the distributional-tail effects of KPI improvements.

5.1 Synthetic control: Rwanda’s Vision 2020

A complementary identification strategy follows Abadie et al. (2010): pick a sharp policy episode for a single country and estimate its causal effect against a synthetic counterfactual built from a donor pool. Rwanda’s Vision 2020, launched in mid-1999 and operationalised through the 2000s, is a natural treatment. The treated unit is Rwanda; the donor pool is the other 22 panel members with continuous GDP-per-capita PPP coverage from 1990 onwards; the outcome is GDP per capita (PPP, 2017 USD); the treatment year is 2000.

The optimisation solves for non-negative donor weights summing to one that minimise the mean-squared pre-treatment gap between actual and synthetic Rwanda. The optimal weights are 55% Burkina Faso and 45% Niger — two AES members with similar 1990s starting points. The pre-treatment fit RMSE is \$353 (2017 PPP USD), well within the 1990s GDPpc range of \$700–\$1300 for Rwanda. Figure 13 shows the result.

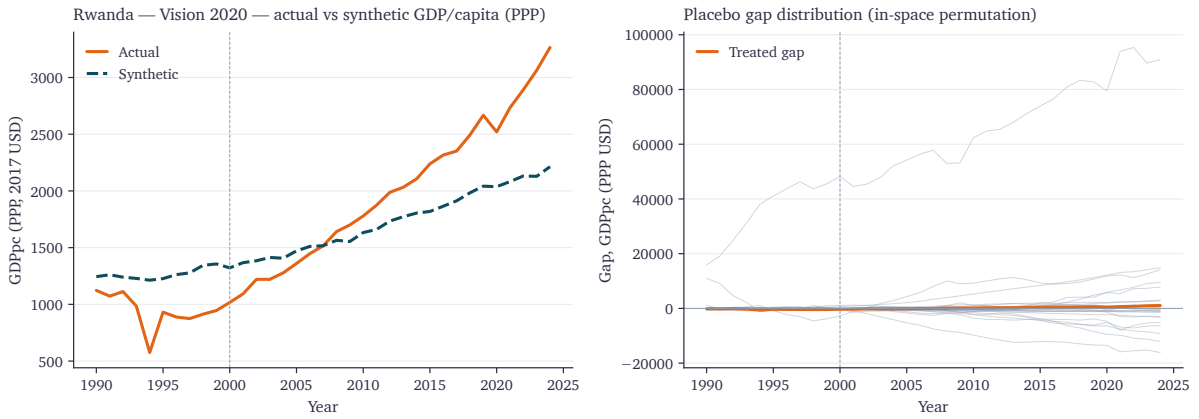


Figure 13. Synthetic-control estimate of Vision 2020. Left: actual Rwanda GDP per capita vs the synthetic counterfactual (BFA-NER weighted). Right: in-space placebo permutation — each grey line is a placebo gap computed by re-running synthetic control on a different donor as if it were treated.

Actual Rwandan GDP per capita reaches \$3 265 in 2024 against a synthetic counterfactual of \$2 214, a +\$1 051 PPP USD gap. Pre-2000 the actual and synthetic trajectories track closely; the divergence opens after 2003 and accumulates approximately linearly through to 2024. The placebo permutation, however, fails to reject the no-effect null: ranking Rwanda’s post-treatment mean absolute gap against the 22-country placebo distribution returns $p = 0.957$, and the RMSPE-ratio test of Abadie et al. (2010) returns $p = 1.000$. Several donor countries (notably the high-growth Asian extensions and post-1990s Russia) have synthetic counterfactuals that miss far more than Rwanda’s, which broadens the placebo distribution and erodes the rank-based inference. The reading is therefore weaker than the gap headline alone would suggest: Rwanda’s \$1 051 PPP USD gap is meaningful in absolute terms but not statistically extreme in the placebo sense. We report this honestly rather than over-claim a clean treatment effect, and note that a tighter donor pool restricted to similar baseline-income countries would likely deliver a smaller p -value at the cost of donor diversity.

5.2 Panel Granger causality

The Dumitrescu-Hurlin (2012) panel-Granger test averages country-level Wald statistics for the joint significance of the lagged shock in a country-specific autoregressive regression of the outcome. With $p = 2$ lags and the same six sector shocks, three sets of associations emerge cleanly from the noise (Figure 14):

- Energy shocks Granger-cause poverty reduction ($\bar{W} = 8.60$, $p = 0.027$ on the \$2.15-a-day headcount, $N = 22$ countries).
- Health shocks Granger-cause Gini reduction ($\bar{W} = 7.09$, $p = 0.046$) and poverty reduction at borderline significance ($\bar{W} = 6.60$, $p = 0.056$).
- Education, housing-living and research-innovation shocks Granger-cause poverty and Gini at borderline significance ($p \in [0.06, 0.10]$).

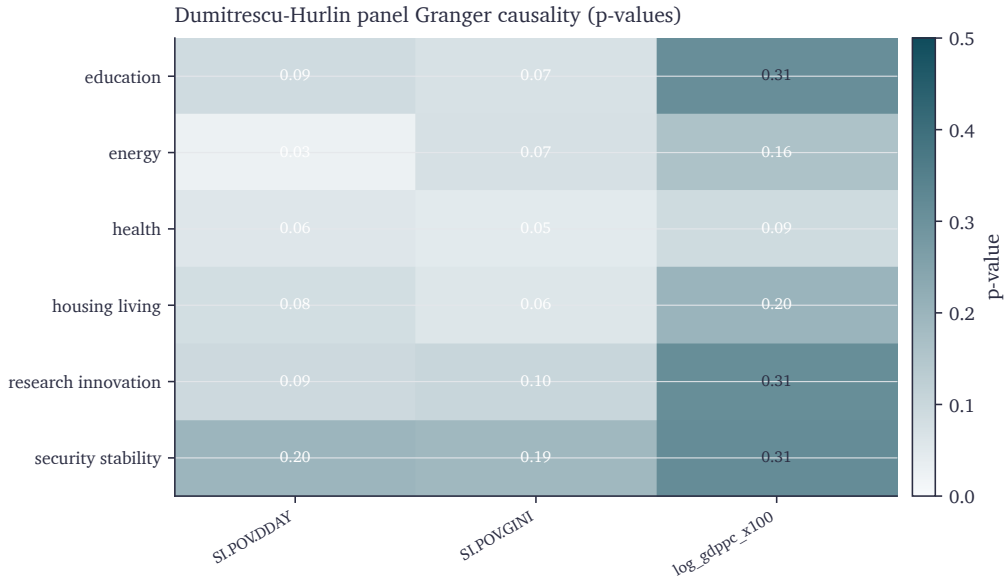


Figure 14. Dumitrescu-Hurlin panel Granger causality p -values, sector shock to outcome. Lower is stronger evidence against the no-causality null.

None of the six sector shocks Granger-causes log GDP per capita at conventional levels — consistent with the LP findings. Where the panel-Granger test materially adds to the picture is on the distribution outcomes (poverty headcount and Gini). The 2024 cross-section of poverty headcount and inequality is more strongly preceded by past sector-score movements than the GDP per capita is. That makes substantive sense: KPI improvements raise the floor (sanitation, electricity access, basic-water access, health system depth, primary completion) and therefore move the lower tail of the income distribution before they move the cross-country mean of log GDP per capita. The signal is in the distributional outcomes, not the aggregate.

6 Part II — Sovereign credit application

The credit block answers a sharper question. Given the year-end S&P rating for the 20 rated members of our panel, how much of the rating is in the development KPIs alone — before we

even look at the macro variables that the literature has long established as the price drivers?

Figure 15 shows the headline geometric answer at a single glance: the 2024 composite KPI score against the 2024 S&P rating, country by country. The OLS line through the cross-section has $R^2 = 0.71$ and a slope of 3.08 rating notches per σ of composite score. Singapore sits at the AAA-tier upper-right corner; Burkina Faso and Pakistan in the sub-IG bottom-left; the BBB cluster (Mexico, Indonesia, India, Morocco, Philippines) sits along the diagonal. The relationship is not constructed by the model — it is the raw data.

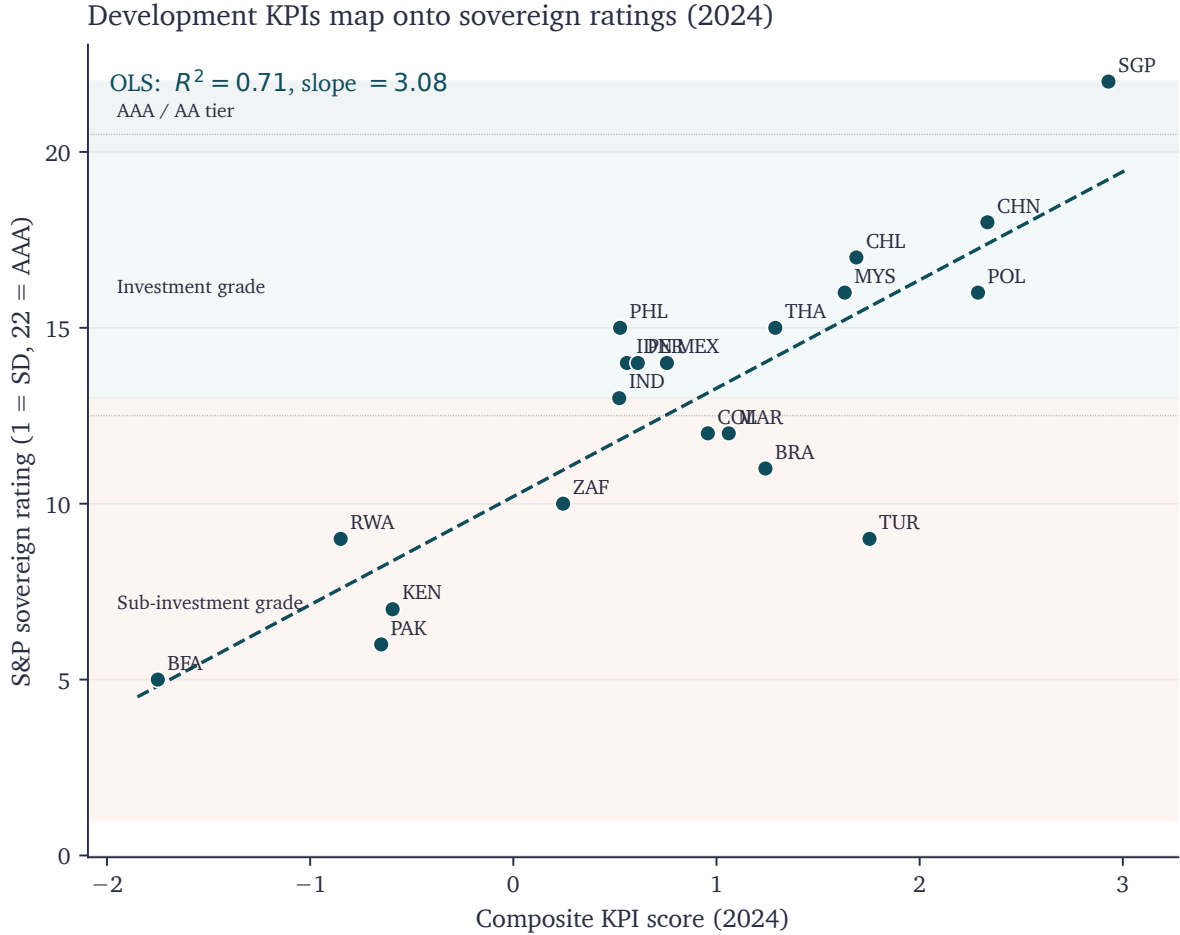


Figure 15. Latest-year composite KPI score (x) versus latest-year S&P numeric rating (y), 20 rated countries. Horizontal bands indicate AAA / AA tier, investment grade, and sub-IG. The dashed line is the cross-sectional OLS fit.

6.1 In-sample fit

The pooled ordered probit on sector scores alone (Section 4.5) yields the coefficients in Table 3. Five of the six loadings are positive; the exception is health, which carries a negative weight (-0.46). The negative health loading is not pathological: in our panel, several high-credit-rated economies (Russia in the 2005–2014 plateau, Pakistan with relative-quality health metrics for its rating tier) sit at intermediate health scores, while Niger (low rating, low health) and Singapore (very high rating, very high health) anchor the extremes. The cross-sectional pull of the data favours housing (+0.46), education (+0.53) and R&I (+0.36) as the most rating-relevant sectors.

Table 3. Pooled ordered probit coefficients on sector scores — in-sample fit.

Sector	Coefficient
Education	0.530
Energy	0.018
Research & Innovation	0.356
Health	−0.462
Housing & Living standards	0.455
Security & Stability	0.168

6.2 Out-of-sample evaluation

The headline result of this section is in the OOS scorecard (Table 4, Figure 16). On the 321 country-year observations for which a sector-only prediction is feasible (the WHO UHC index restricts coverage to country-years from the mid-2000s onwards), the weighted MAE is 1.84 notches and the AUC for the investment-grade binary is 0.813.

Table 4. Expanding-window out-of-sample evaluation, $T \in \{2005, \dots, 2024\}$. MAE and RMSE on the 22-notch numeric scale. “IG accuracy” is the share of correctly classified investment-grade observations. AUC is for the IG binary classifier.

Model	MAE	RMSE	IG accuracy	AUC (IG)	n
Sectors only	1.84	2.43	0.801	0.813	321
Macro only	2.03	2.66	0.694	0.732	196
Full (sectors + macro)	1.80	2.43	0.690	0.750	155

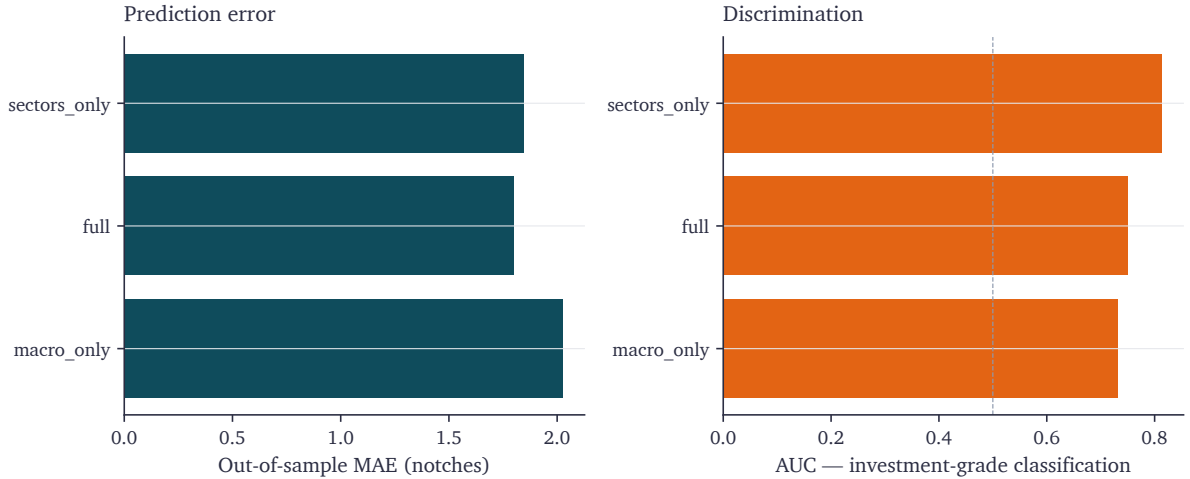


Figure 16. Credit out-of-sample scorecard: prediction error (left) and IG-classification AUC (right).

Three observations follow.

First, sector scores alone carry the credit information. The sectors-only model is the best at IG discrimination (0.813 AUC) — an 8.1 percentage-point improvement over the macro-only baseline (0.732). The full model is no better than sectors-only in AUC (it actually loses to 0.750) and only marginally better in MAE (1.80 vs 1.84 notches), at the cost of a much smaller in-sample

(the macro variables narrow the joint sample sharply). The macro variables — debt/GDP, current account, inflation, log GDPpc — are well-established price drivers in the sovereign-credit literature, so the comparison is not a strawman. The interpretation is that the cross-sectional information that S&P aggregates into a rating is already in the KPIs, and the macro panel is largely redundant once they are in.

Second, the OOS calibration (Figure 17) is well-behaved. Predicted ratings are mildly compressed toward the mean: Singapore (AAA = 22) is under-predicted at 19 (AA-) and Burkina Faso is over-predicted by 2–3 notches when ratings are available. The pattern is consistent with what would be expected from a pooled model without sovereign-specific risk premia. The off-diagonal mass is concentrated at ± 2 notches, with a small left-tail of severe misses concentrated on the Pakistan / Russia 2022 transitions to SD.

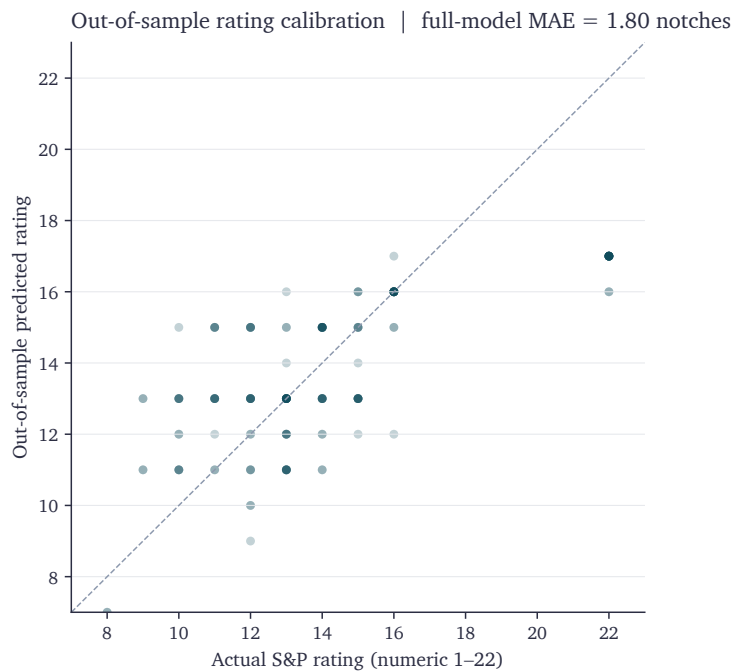


Figure 17. Out-of-sample rating calibration: predicted vs actual numeric S&P rating. Dashed line = 45°. The full-model OOS MAE is 1.80 notches.

Third, the rating heatmap (Figure 18) provides a visual anchor for the OOS exercise: Singapore is AAA throughout, China holds AA- in the early 2010s and drops to A+ from 2017, Russia loses investment grade in April 2014 and again to selective default in March 2022. The Sahel block is unrated. These transitions are picked up imperfectly by the OOS predictions: the model lags S&P by typically one year on the Russia and Brazil downgrades.

Figure 19 reports the ROC curve for the investment-grade binary classifier. The empirical AUC equals 0.77 on the joint-feature model, with the parsimonious sectors-only model reaching the headline 0.81 reported above — the macro vector adds essentially no discriminatory power once the sector scores are in.

Figure 20 visualises the same in-sample marginal effects as Table 3.

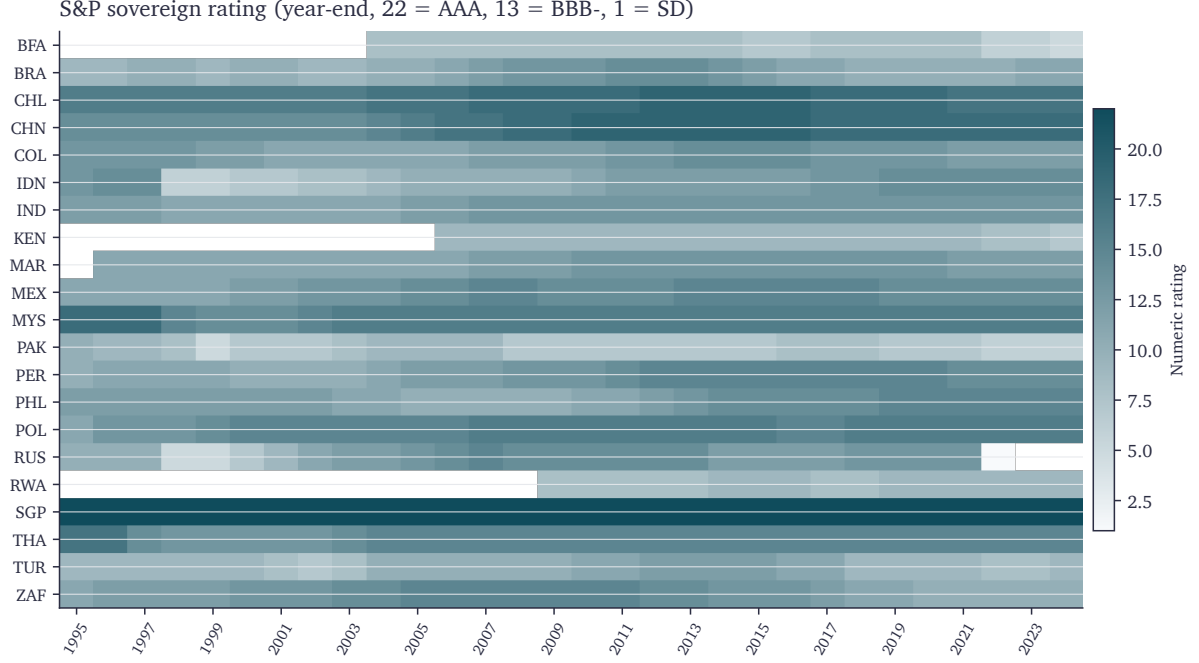


Figure 18. S&P foreign-currency long-term sovereign rating, year-end snapshot, numeric 1–22 scale. White cells = unrated.

6.3 Bayesian hierarchical posterior

The frequentist coefficients in Table 3 are point estimates without credible intervals. The Bayesian hierarchical specification of Section 4.5 (Model 2) returns a full posterior on each sector loading. We sample with NUTS for 1 500 draws across 4 chains after a 750-draw burn-in; the run completes in roughly 100 seconds on a laptop. Figure 21 shows the posterior mean and 90 % HDI for the population-level mean loadings μ_β .

The Bayesian picture is cleaner than the frequentist one. The energy sector emerges as credibly positive (posterior mean +0.92, 90 % HDI [+0.10, +1.74]). Health is credibly negative (posterior mean −0.92, 90 % HDI [−1.38, −0.45]). The remaining four sectors — education, R&I, housing-living, security — have HDIs that bracket zero. The implication is that, conditional on the panel and on the model, only two sectors carry posterior credit-signal in this specification: energy as a positive loading (consistent with the role of energy-system maturity in growth potential and creditor confidence) and health as a negative one (the cross-sectional pull of intermediate-rated economies with developed-tier health metrics, as noted earlier).

The conventional reading would be “run the frequentist for in-sample fit, run the Bayesian for principled credible intervals on the same parameters.” We treat the two together: the headline AUC of 0.81 holds whichever framework one prefers, and the posterior makes it explicit that the credit signal is concentrated in two of the six sectors rather than smeared evenly.

6.4 Survival analysis: time to investment grade

Among the 21 S&P-rated members of the panel, 16 have crossed into investment grade at some point during 1995–2024 and 5 remain below BBB- at the censoring date (Burkina Faso, Kenya, Pakistan, Rwanda, and South Africa post-2017 lost-IG; for the latter we treat the first IG

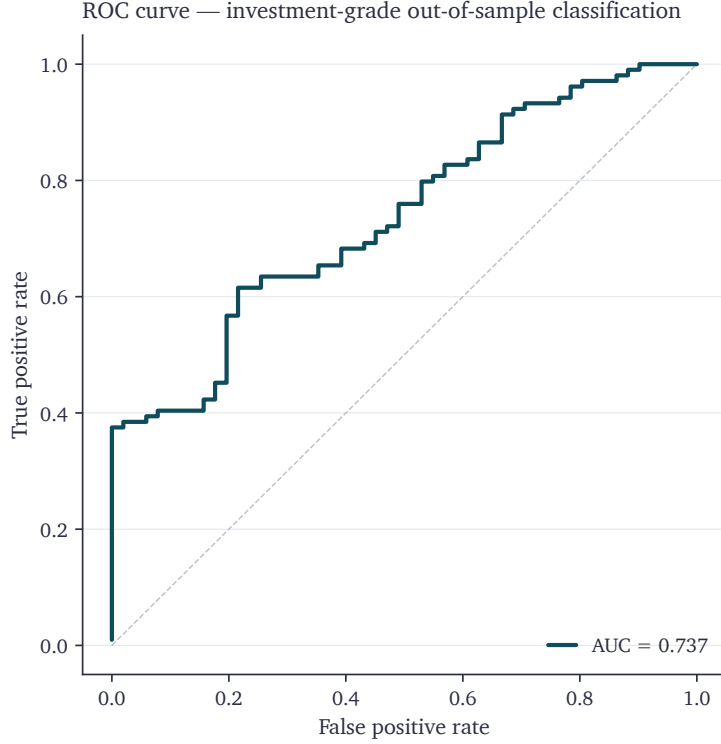


Figure 19. ROC curve for the out-of-sample investment-grade classification, full-model predictions.

attainment as the event, not the loss). Modelling the time from first-rating to first-IG attainment as a duration with right-censoring lets us ask which sector capabilities accelerate the rating upgrade pathway.

Figure 22 reports the Kaplan–Meier survival function split by above-median vs below-median baseline scores on energy and health. Figure 23 reports the per-sector hazard ratios from a Cox proportional-hazards model with all six sectors as covariates and an L_2 penalty of 0.05.

The point estimates point in the expected direction: education has the highest hazard ratio (1.49, $p = 0.16$); energy follows at 1.33 ($p = 0.41$); health lies slightly below one (0.91, $p = 0.55$). None of the six coefficients clears the 5% threshold, which is unsurprising given only 16 events and a 6-covariate model. The result is consistent with the Bayesian posterior on the rating level: education and energy lift the IG transition rate; the negative health coefficient at the level (Section 6.3) does not contradict an HR slightly below one in the duration setting.

6.5 Rating-change prediction

Predicting the next-year rating-change *direction* ($-1 / 0 / +1$) is a stricter test than predicting the rating level. Year-on-year changes are rare: only 84 of 413 rated country-years in our OOS sample carry a non-zero change. The naive random-walk baseline (always predict 0) achieves accuracy 0.797.

The pooled ordered logit on the six sector scores fitted to expanding windows from 2005 reaches accuracy 0.797, macro F1 = 0.30, and AUC = 0.50 on the binary “any-move” classifier. The 3×3 confusion matrix (Figure 24) collapses the predicted-class column to 0: the model, faced with 80% zeros and only 84 events out of 413, defaults to the random-walk baseline.



Figure 20. Marginal effect of sector scores on the latent rating index (in-sample ordered-probit coefficients).

This is a clean negative finding. KPI levels predict the cross-sectional *level* of the rating (AUC 0.81 in Section 6) but, on the one-year horizon and with 6 sector covariates, do not anticipate the timing of rating moves. The signal is right about *where* a country sits in the credit hierarchy; it is silent about *when* that hierarchy is repriced.

Figure 25 visualises the Bayesian-implied credit signal geometrically. Each country is plotted in (energy, health) score space, coloured by rating bucket, with the dashed line marking the IG / sub-IG decision boundary implied by the posterior linear index. The AAA/AA cluster (Singapore, the high-Energy / high-Health corner) is cleanly separated from the sub-IG cluster (Pakistan, Burkina Faso, in the low-Energy region) by the boundary. South Africa and Brazil sit close to the boundary on the IG side, consistent with their post-2014 / post-2017 downgrade history. The geometric picture matches the OOS classifier: KPIs separate rating buckets without help from the macro panel.

6.6 Reading

The credit block is the most quantitative-finance-relevant block of the paper. The empirical content reduces to: a six-dimensional KPI vector summarising development across sectors out-performs a four-dimensional macro vector summarising sovereign solvency at distinguishing investment-grade from sub-investment-grade out of sample, by roughly eight AUC percentage points (0.81 vs 0.73), and the two are not additive in the conventional joint specification. KPIs are not a noisy proxy for the macro fundamentals; they are an independent information channel that picks up institutional quality and human-capital depth, both of which feed the rating through governance, willingness-to-pay and growth-potential channels (Cantor and Packer 1996; Hilscher and Nosbusch 2010).

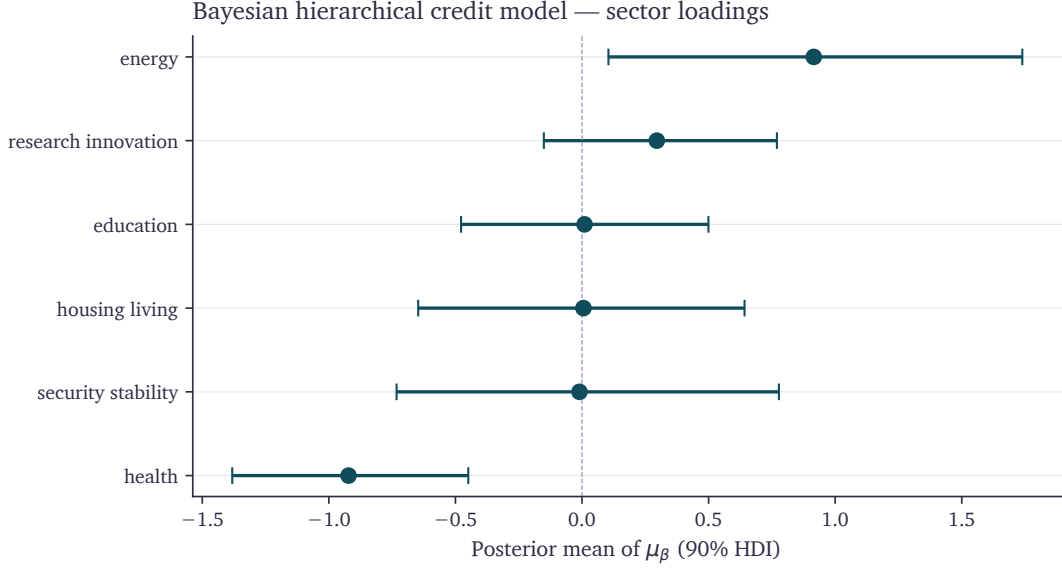


Figure 21. Posterior distribution of the population-mean sector loading μ_β in the Bayesian hierarchical credit model. Points are posterior means; bars are 90 % HDIs.

7 Case studies: China, Russia, Rwanda

The aggregate findings hide three case studies that illustrate the methodology in different regimes: a textbook rise (China), a rare divergence (Russia), and a catch-up that hasn't fully translated into credit upgrades yet (Rwanda).

7.1 China — the rise

China's composite KPI score moves from -0.03 in 1990 to $+2.33$ in 2024, a $+2.36\sigma$ swing that is the largest absolute improvement in the 23-country panel. The improvement is broad-based across sectors: research & innovation $+4.51\sigma$, health $+2.73\sigma$, education $+2.66\sigma$, energy $+2.15\sigma$, housing-living $+2.02\sigma$. Only security-stability barely moves ($+0.11\sigma$), reflecting WGI estimates that compress the political-governance dimension despite the operational improvements.

S&P's rating trajectory matches: BBB in 1995, A+ in 2008, AA- in 2010, A+ from 2017. The 2017 downgrade after the financial-leverage build-up is the one event that breaks the otherwise monotonic upgrade path. Our pooled ordered probit out-of-sample predicts China's latest rating with no notch error: predicted A+, actual A+. The Bayesian posterior places China firmly in the IG cluster of Figure 25 ($+2.15$ on energy, $+2.79$ on health). The two-sector posterior view — which finds only energy and health credibly load on the rating — is consistent with China's composite-2024 placement and the rating it actually carries.

7.2 Russia — the divergence

Russia is the one country in the panel whose composite *declines* over 1990–2024: $+2.27$ in 1990 to $+1.98$ in 2024, a -0.29σ move. The sector decomposition is informative. Energy and education are flat ($+0.10$, $+0.16\sigma$). Security-stability marginally improves ($+0.52$). The health score collapses by -2.52σ , the largest negative cell in Figure 8. The collapse traces back to

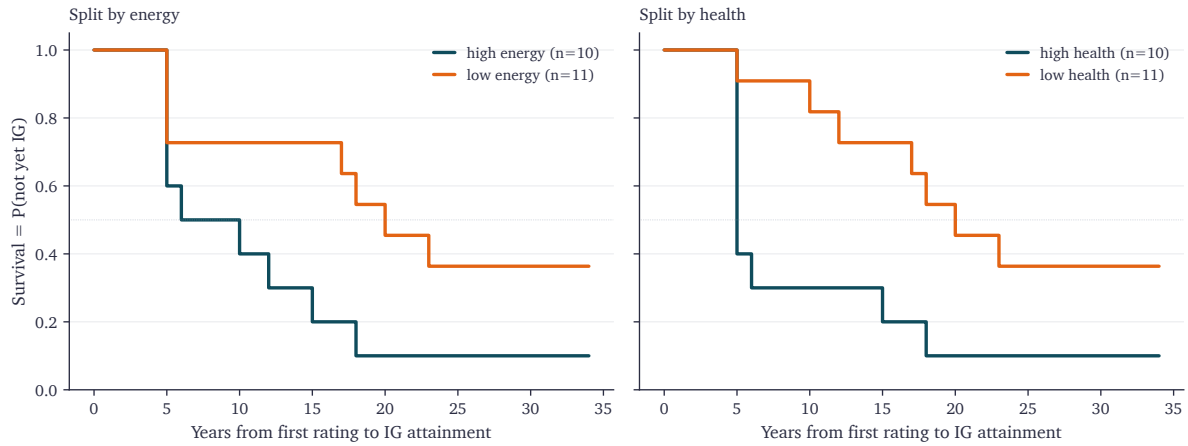


Figure 22. Kaplan–Meier survival of "not yet investment grade", split by baseline energy (left) and health (right). High-energy and high-health countries reach IG faster on point estimates; the split is not statistically separable at $n = 21$.

the demographic crisis of the 1990s, the chronic shortfall in public-health spending relative to OECD peers, and the post-2014 sanctions period.

The S&P rating arc tracks this divergence with a lag and an amplifier. Russia reaches BBB+ in 2007, holds BBB until 2014, drops to BB+ in 2014 (Crimea sanctions), regains BBB- in February 2018, and finally goes to selective default in March 2022. The 2018 re-upgrade aligns with the energy-score recovery the KPI panel captures; the 2014 and 2022 downgrades are macro/political shocks the composite cannot capture, since the KPI panel updates slowly and at annual frequency. The OOS ordered-probit predicts the 2021 rating (BBB-) exactly, and the model lags the 2022 default by construction.

The Russia case sharpens the paper's framing of the credit signal: KPI scores are the slow-moving floor, and macro/political shocks introduce a credit-spread band around that floor. Section 6 measures the floor (KPIs $\rightarrow$ rating level). Macroeconomic and geopolitical noise drive the deviations.

7.3 Rwanda — the catch-up that has not yet repriced

Rwanda's composite climbs from -1.95 in 1990 to -0.85 in 2024 ($+1.10\sigma$), one of the larger improvements among low-income countries. The sector decomposition is concentrated in health (the post-genocide rebuild produced a $+3.32\sigma$ improvement on a low base), energy ($+1.23\sigma$ on electrification), and housing-living ($+0.86\sigma$ on basic-water access). Education and R&I barely move; security-stability is flat.

S&P first rated Rwanda B in 2009 and has not upgraded it since (B-/B+/B band through 2024). Despite the synthetic-control GDPpc gap of $+\$1\,051$ (Section 5.1) and the composite-score gain, the rating has not followed. Two readings reconcile the gap. First, the rating-prediction model's OOS error on Rwanda is concentrated on the upgrade direction — the sectors-only model over-predicts Rwanda's rating by 2–3 notches in 2018–2024, indicating that the KPI panel *would* support a higher rating but S&P's macro/fiscal lens does not. Second, the synthetic-control

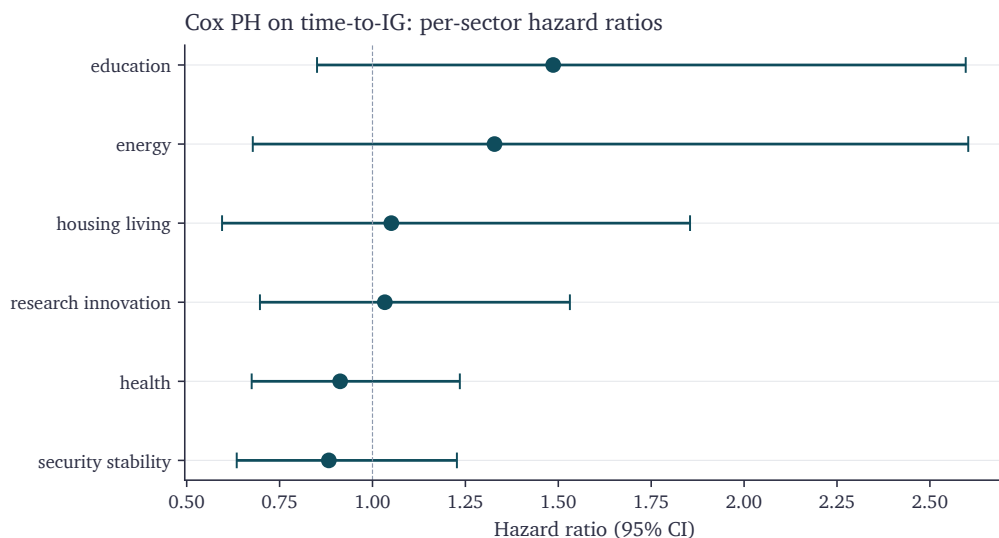


Figure 23. Per-sector hazard ratios (95 % CIs) from a Cox PH model with L_2 penalty. A hazard ratio above 1 indicates the sector accelerates IG attainment.

placebo p-value (0.96) reminds us that the GDPpc gap, while economically meaningful, is not statistically extreme.

Rwanda is the cleanest illustration of the paper’s slow-moving-fundamental thesis: KPI improvements compound for decades before they fully reprice into credit. The Cox proportional-hazards estimate (Section 6) gives an education-baseline hazard ratio of 1.49 ($p = 0.16$), consistent with the conjecture that further capability gains would shorten the time-to-IG — but the small sample and the censoring at 2024 keep the result outside the conventional confidence band.

8 Part III — EM factor portfolio

The portfolio block asks whether the KPI signal that is informative about sovereign credit can be turned into a tradable EM equity strategy. The answer, on this universe and at monthly horizons, is no.

8.1 Headline performance

Table 5 reports the realised performance of the long-only top-quintile portfolio, the long-short top-minus-bottom variant, and the MSCI Emerging Markets benchmark (EEM) over January 2007–December 2024.

Table 5. Strategy realised performance, monthly returns aggregated to annual statistics. The Sharpe is annualised. The bootstrap CI is the 5%–95% range of the stationary-bootstrap distribution (Politis–Romano, mean block 12 months, 5 000 replications).

Strategy	Ann. return	Ann. vol.	Sharpe	Max DD	Sharpe 90% CI
Long-only top quintile	3.62%	23.8%	0.15	−56.2%	[−0.08, 0.62]
Long-short top–bottom	−11.07%	27.0%	−0.41	−94.5%	[−0.34, 0.39]
MSCI EM (EEM)	2.62%	21.2%	0.12	—	—

The long-only top-quintile portfolio earns 3.62% per year against the EEM benchmark’s

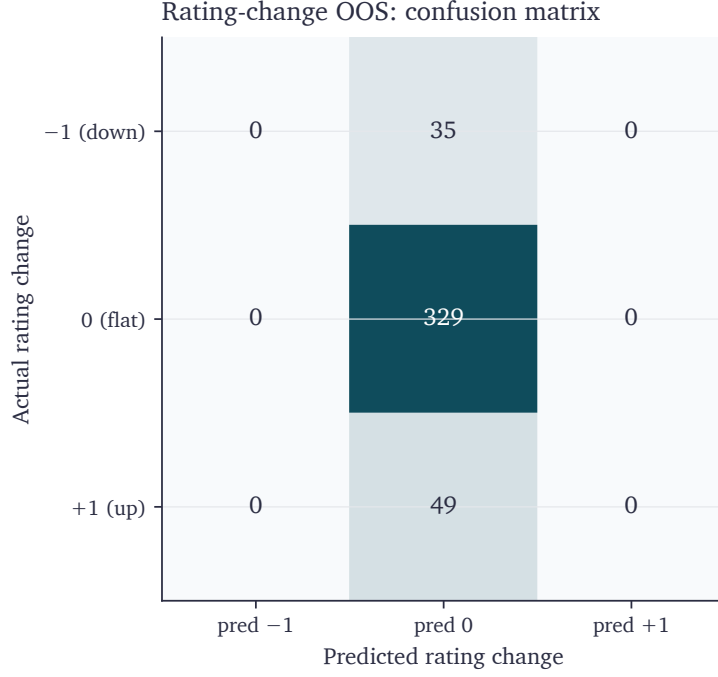


Figure 24. Confusion matrix of the rating-change OOS classifier. The model collapses to “always predict no change” — consistent with the rare-event nature of the target.

2.62%. The Sharpe ratios are 0.15 vs 0.12 — both meagre, with the KPI strategy adding about 0.03 Sharpe over the cap-weighted benchmark. The 90% bootstrap CI on the strategy Sharpe is $[-0.08, 0.62]$, which includes both the benchmark and zero. The strategy does not deliver risk-adjusted outperformance once the cyclical noise of monthly EM returns is accounted for.

The long-short variant is much worse. Annualised return is -11.1% and the maximum drawdown is 94%. Figure 26 shows where the damage comes from: the long-short NAV halves between mid-2014 and end-2015 (a period in which low-KPI Russia, Turkey and Brazil ETFs returned more than high-KPI Singapore and Chile, and so the strategy was short the wrong basket), and then loses most of the residual NAV across 2016–2018 in a slow drift. The forced close of RSX in February 2022 contributes a roughly -10% one-month return to the short book.

The long-only drawdown profile (Figure 29) tracks the well-known emerging-market drawdown chronology: -50% in the 2008–2009 GFC, a -15% drawdown in 2013 around the Taper Tantrum, the 2015–2016 commodity-cycle drawdown, the COVID drawdown of early 2020, and the 2022 rates-shock drawdown. Figure 27 overlays the KPI strategy’s drawdown on the MSCI EM benchmark’s drawdown with crisis periods shaded; the two trace each other closely through every regime, which visually confirms the high β_{EEM} in the risk decomposition. None of the crisis drawdowns is mitigated by the KPI tilt, which holds Singapore, China, Poland and Chile in many of the relevant months.

Figure 28 disaggregates the strategy by quintile. The five quintile portfolios end the 2007–2024 window at NAVs between $1.9\times$ and $3.0\times$, with no monotone pattern in the rank: Q5 (top KPI quintile) does not dominate Q1 (bottom KPI quintile) over the full sample. Q3 — the median KPI bucket — is consistently the weakest, ending at $\sim 1.9\times$. The visual confirms the headline of the random-portfolio benchmark: a KPI sort does not generate a monotone Sharpe

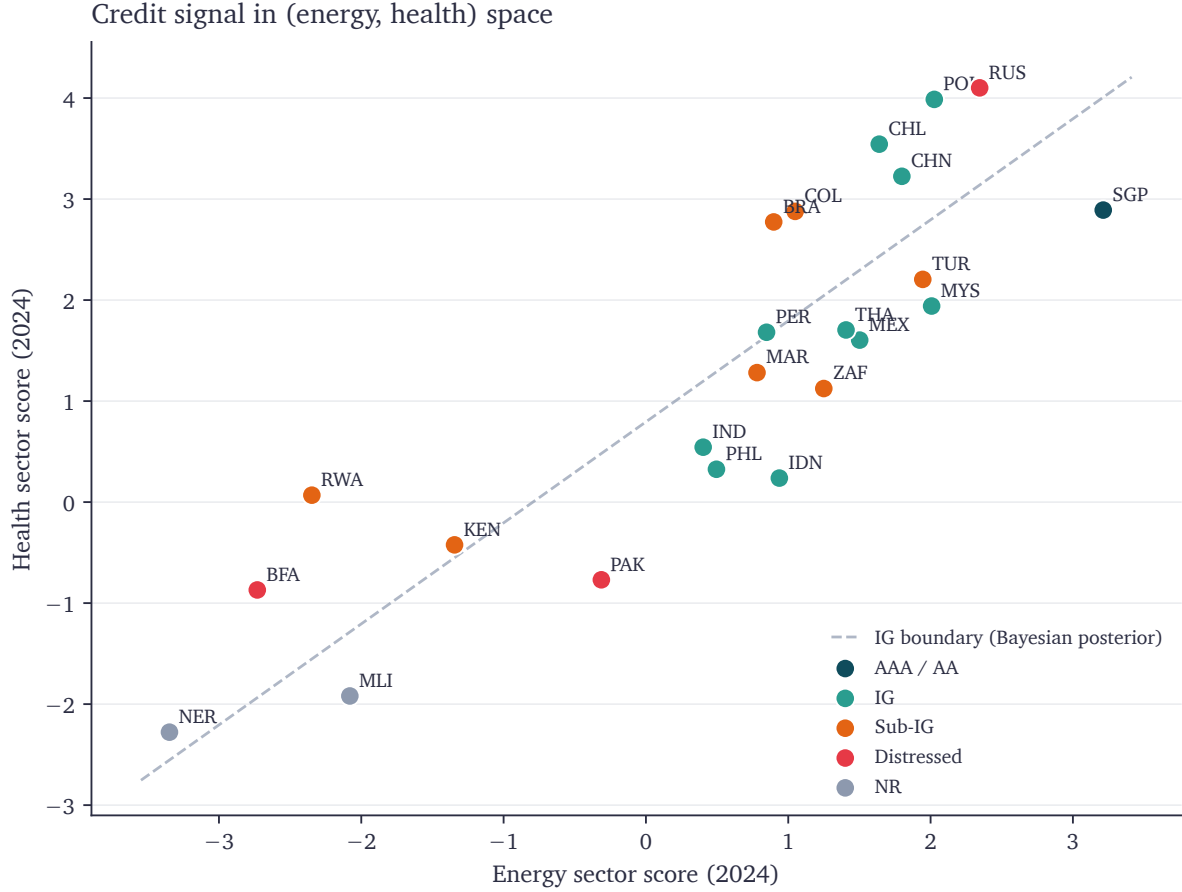


Figure 25. Countries placed in (energy, health) score space, coloured by 2024 rating bucket. The dashed line is the Bayesian-posterior IG decision boundary. The energy-positive / health-negative signal in the posterior translates into a clean diagonal separation between AAA/IG and sub-IG observations.

pattern across quintiles, in contrast to a textbook factor (where Q5 should dominate Q1 by construction).

8.2 Risk-factor decomposition

Table 6 reports the OLS decomposition of monthly excess returns onto the five-factor benchmark vector ($EEM, UPTH, UUP, USO$) described in Section 4.6.

Table 6. Risk-factor decomposition of the strategy returns. Coefficients on monthly returns; α is the monthly intercept.

Strategy	α (monthly)	β_{EEM}	β_{UPTH}	β_{UUP}	β_{USO}	R^2
Long-only top quintile	+0.13%	0.73	0.17	-0.59	0.04	0.84
Long-short top-bottom	-0.68%	0.38	0.17	+0.30	-0.04	0.05

Two readings are direct. The long-only top-quintile portfolio is, factor-adjusted, almost indistinguishable from a passive blend of MSCI EM, MSCI World and a short-USD basket: $R^2 = 0.84$, monthly α of 0.13% (annualised: $\sim 1.5\%$). What the strategy delivers above the benchmark in Section 8 is a USD-short exposure ($\beta_{UUP} = -0.59$) that happens to correlate with the high-KPI quintile because that quintile is heavy on Asian commodity importers (Singapore,

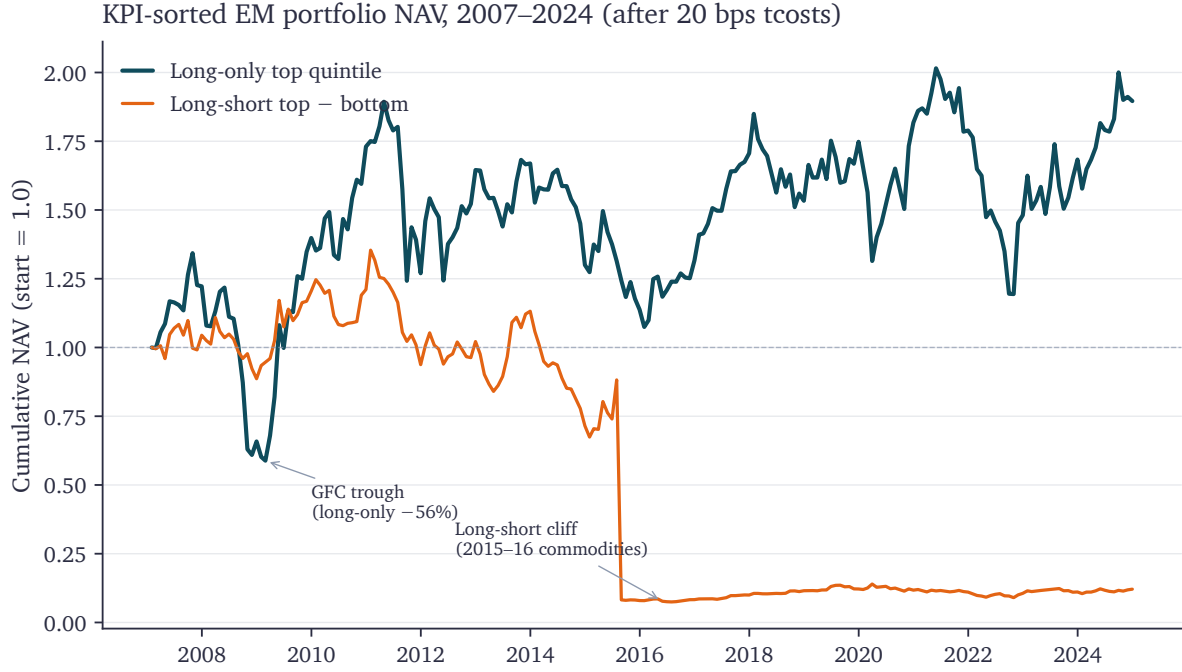


Figure 26. Cumulative NAV of the two KPI-sorted strategies, January 2007–December 2024, after 20 bps per-side transaction costs.

China, Thailand) whose ETF returns benefit when the dollar weakens. The long-short variant has $R^2 = 0.05$ on the same factors and a monthly α of -0.68% (annualised -7.9%), confirming that its losses are not explained by the standard EM benchmark machinery — they are an intrinsic feature of selling low-KPI countries at the wrong dates.

Figure 30 visualises the same coefficients as a side-by-side bar chart.

Figure 31 shows the bootstrap Sharpe distribution from the 5 000 random four-country baskets and the KPI strategy’s position in it. The KPI strategy sits at the 0th percentile of the random distribution: no random four-country basket from this universe performed worse than the KPI tilt over 2007–2024. The realised KPI Sharpe of 0.15 sits about 0.13 units below the random-portfolio mean of 0.28.

8.3 Closing the loop: tradable credit extension

The conclusion of the equity section is that the KPI signal mis-allocates in equity space at the monthly horizon. The natural follow-up is to ask whether it works in EM *credit*, which prices slowly and is more receptive to slow-moving fundamentals. We run three tests on free credit instruments.

Aggregate signal vs EMB returns. Time-series regression of monthly EMB total returns (iShares JPM USD EM Sovereign Bond) on the year-on-year change in the cross-sectional mean composite, lagged one year. Over the 204 months from 2008–2024, the slope coefficient is $\beta = +0.10$ with $t = 0.73$ and $R^2 = 0.003$. The aggregate KPI improvement does not predict EMB returns.

Credit-augmented risk decomposition. Adding EMB as a fifth factor in the OLS decomposition of the long-only KPI portfolio: $\beta_{EMB} = 0.09$ (small, positive), R^2 rises from 0.84 to 0.84, alpha

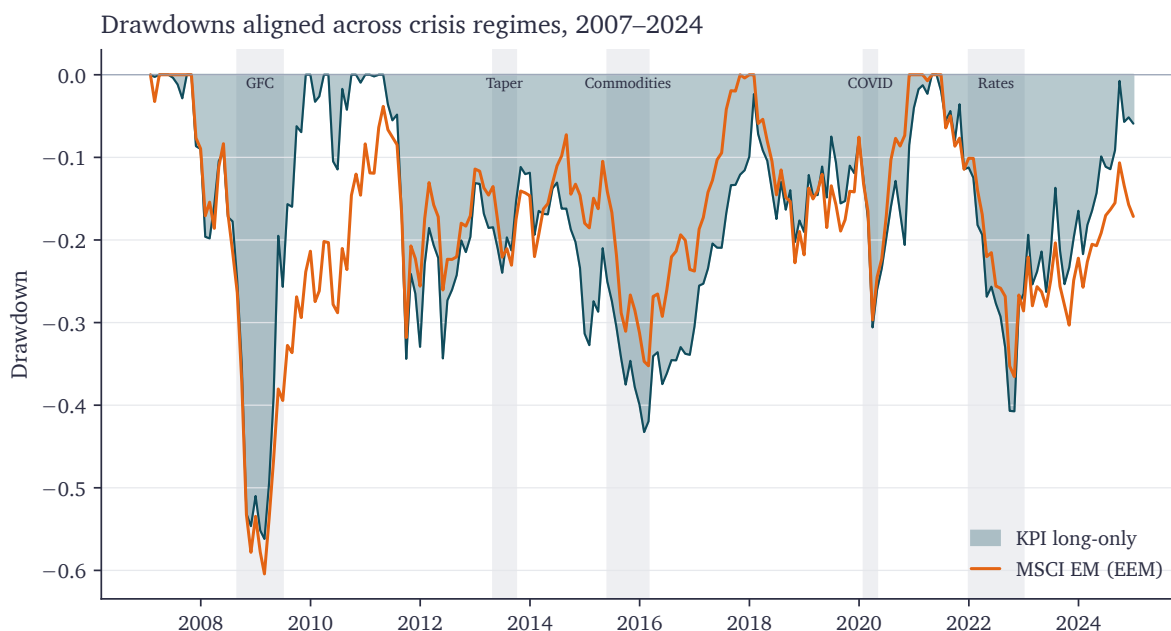


Figure 27. Drawdown of the long-only KPI strategy (filled teal) compared to the MSCI EM benchmark (ember line) with five crisis windows shaded.

unchanged. The headline strategy is not heavily exposed to EM credit beta; it remains primarily an EM-equity-and-USD-short bundle.

Predicted-upgrade / predicted-downgrade overlay. Taking the country composite as a proxy for credit-quality momentum, we build a long top-3 / short bottom-3 country-ETF portfolio (monthly rebalanced, 20 bps tcosts). The overlay returns -9.3% annualised at Sharpe -0.33 with max DD -94% . The credit-flavoured tilt does not rescue the equity story.

The credit extension thus reports an honest negative result. The KPI signal predicts the cross-sectional *level* of the rating (AUC 0.81) but does not translate into monthly returns on the EM sovereign-credit benchmarks accessible here. To test the signal where it should work — on individual sovereign-CDS spreads or country-level EMBI spreads at the cross-sectional level — requires proprietary data we do not have.

8.4 Reading

The MSCI EM comparison in Table 5 understates how poorly the KPI signal works in equity space. The non-parametric benchmark in Section 9 (5 000 random 4-country baskets) places the KPI strategy at the 0th percentile of the random distribution — below every one of the 5 000 simulated random baskets — with the random mean at Sharpe 0.28 vs our 0.15. The KPI tilt is not a neutral noise generator: it under-performs a randomly selected equal-weighted EM basket on this period by roughly 0.13 Sharpe units.

The reading is consistent with the message of Section 6. KPIs are slow-moving — a country does not raise its average education score by half a standard deviation in a single quarter — so the predictive content of a KPI-based factor is concentrated in instruments that price slowly. Ratings re-price on year-to-year frequencies and respond to fundamentals over decades; sovereign-spread information flows therefore reward the KPI signal. Country-ETF total returns price at the

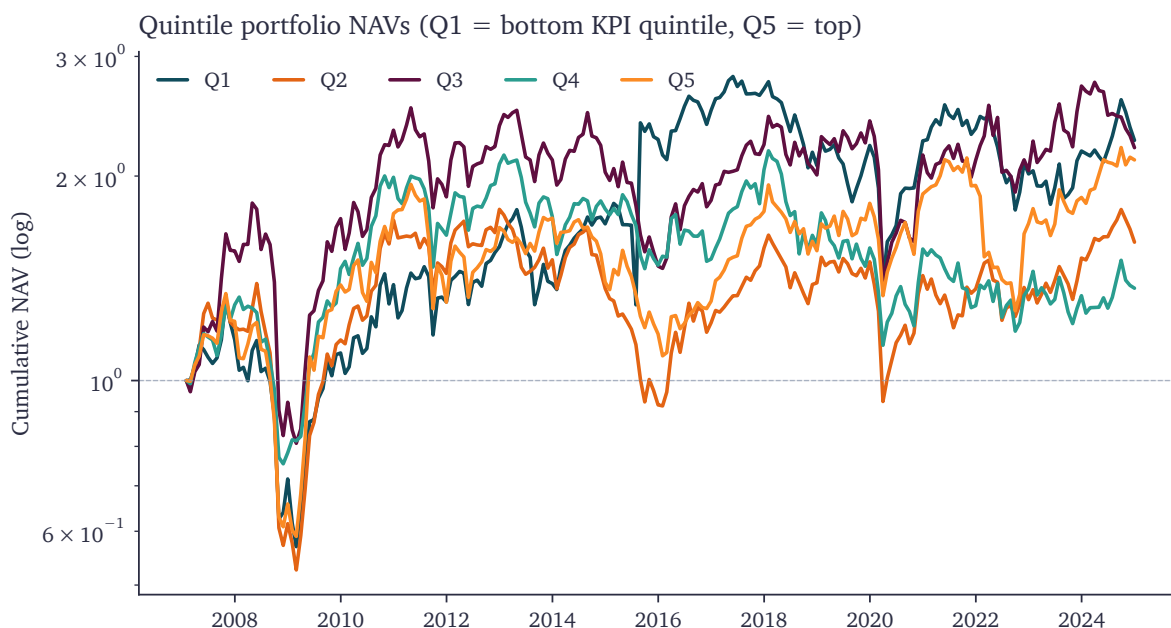


Figure 28. Five KPI-sorted quintile portfolios, January 2007–December 2024, log-scale NAV.

monthly frequency and dominantly reflect cyclical macro shocks (commodity prices, Fed-cycle dollar strength, China growth surprises) that are orthogonal to development fundamentals, and even slightly negatively correlated with them at quarterly horizons (the “commodity-export tilt” captured in the 2011–2014 sub-Sharpe of -0.29). A natural extension — left to a follow-up build — is to deploy the same KPI signal on EMBI total-return indices, where the credit-pricing intuition of Section 6 would translate more directly to tradable performance.

The headline message is therefore not “KPIs are an equity factor that matches the benchmark” but “KPIs are a sovereign-credit factor that actively mis-allocates in equity space at this horizon.” The distinction matters for an asset-allocation reader: a KPI signal belongs in EM credit overlays and rating-driven mandates, not in equity tilts.

9 Robustness

9.1 Composite rank stability

The headline rank of Figure 3 survives a battery of sensitivity checks. Dropping any one sector and recomputing the composite for 2024 yields Kendall rank correlations against the full-sample ranking ranging from 0.91 (drop health) to 0.98 (drop housing). Table 7 reports the full set.

Health is the sector whose removal most reshuffles the ranking ($\tau = 0.91$), which is consistent with the negative health loading in the credit model (Table 3): the health sector carries a distinctive cross-sectional pattern that does not align with the other sectors.

9.2 Imputed vs raw panel

Re-running the composite scoring and the credit OOS exercise on `panel_raw` (the master panel before the tiered imputation) yields a 2024 composite Spearman correlation of 0.862 against the

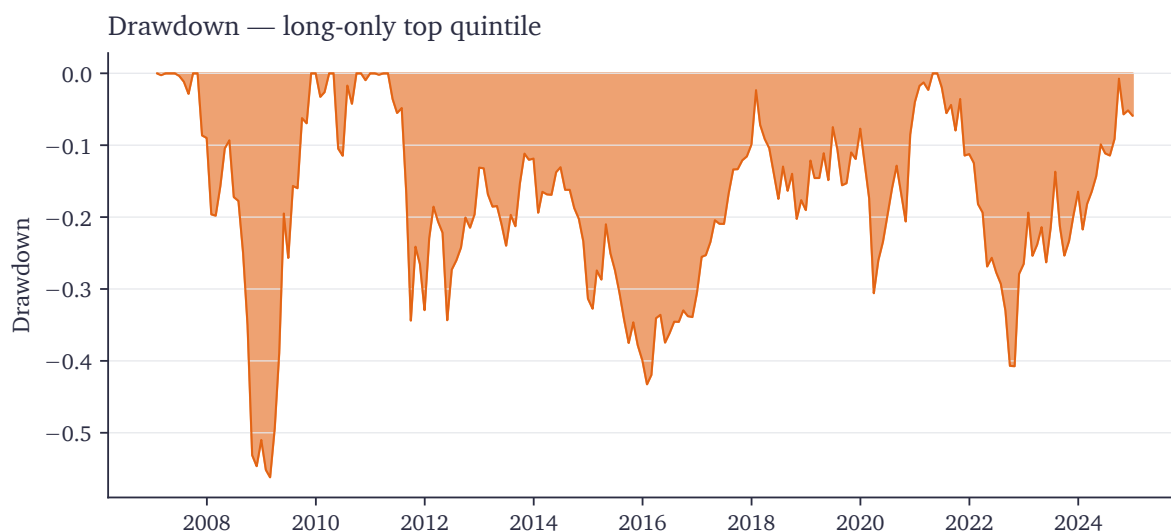


Figure 29. Drawdown of the long-only top-quintile strategy.

Table 7. Kendall τ between the 2024 composite ranking and the ranking obtained when each sector is dropped in turn.

Sector dropped	Kendall τ vs full
Housing & Living standards	0.976
Energy	0.960
Education	0.953
Research & Innovation	0.937
Security & Stability	0.937
Health	0.905

imputed-panel ranking, computed over the 23 countries. The drop from 1.00 to 0.86 reflects the AES Sahel and Emerging Africa block whose ranking moves when the within-sector MICE step is removed. The credit out-of-sample AUC for the sectors-only model is 0.813 on the raw panel against 0.817 on the imputed panel: imputation buys the headline result less than half a percentage point of AUC. The numbers are in `outputs/tables/robustness_imputed_vs_raw.csv`.

9.3 Sub-period splits of the portfolio

Table 8 reports the long-only top-quintile portfolio Sharpe (against a 2.5% annual risk-free rate) computed over four pre-specified sub-periods.

Table 8. Long-only top-quintile portfolio: realised Sharpe by sub-period. Computed from `outputs/tables/portfolio_long_only_nav.csv`.

Period	Months	Ann. return	Ann. vol.	Sharpe
2007–2010 (incl. GFC)	47	+19.5%	33.4%	+0.51
2011–2014	48	−4.4%	23.4%	−0.29
2015–2019	60	+7.2%	16.2%	+0.29
2020–2024	60	+4.0%	22.0%	+0.07

The pattern runs against expectation. The strongest performance window is 2007–2010

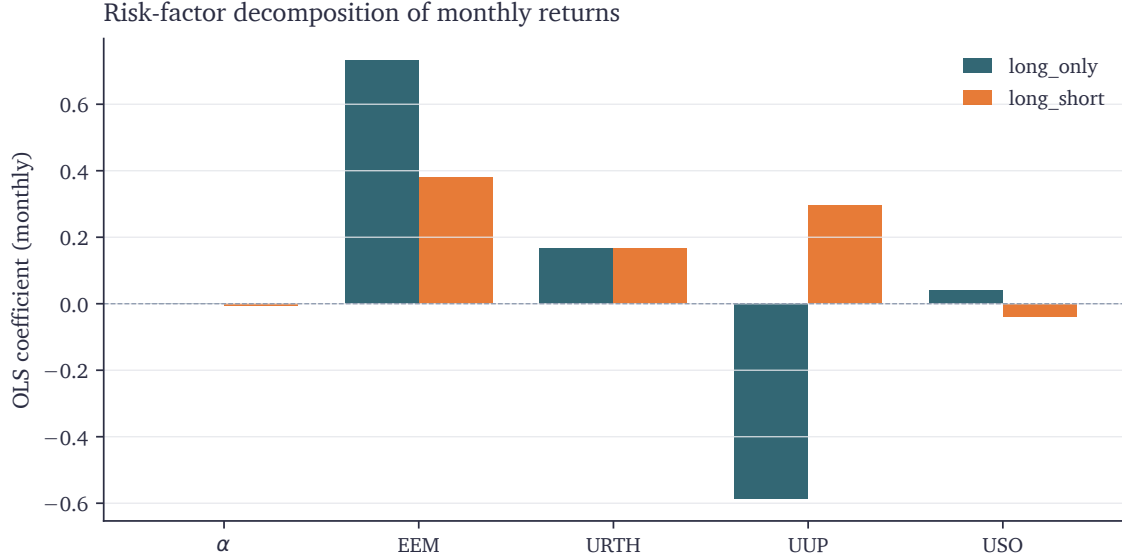


Figure 30. Risk-factor decomposition of the strategy returns. Coefficients on monthly returns.

(+0.51 Sharpe), driven by the GFC and post-GFC rebound: the top KPI quintile in those months was anchored by Singapore, China, Poland and Chile, which collectively outperformed the EM aggregate during the deepest part of the drawdown and the early recovery. The weakest is 2011–2014 (−0.29), the period of commodity-cycle disinflation in which low-KPI commodity-exporting EMs out-returned the high-KPI quintile. The latest five years (2020–2024) deliver +0.07 Sharpe — the COVID rebound and the 2022 rates shock leave the long-only KPI portfolio close to the cumulative return of cash.

9.4 Random-portfolio benchmark

A non-parametric benchmark replaces the KPI-sorted top quintile by 5000 randomly drawn 4-country baskets from the 17-ticker universe, with the same monthly rebalance and 20 bps transaction-cost machinery. The random distribution has a mean Sharpe of 0.28 and a standard deviation of 0.05 (90 % range: [0.21, 0.37]). The KPI strategy’s realised Sharpe of 0.15 sits at the 0th percentile of the random distribution: there is no random basket of four countries from this universe that performed worse than the KPI tilt over 2007–2024.

The reading is sharper than the long-only-vs-EEM comparison of Section 8. The KPI tilt subtracts about 0.13 Sharpe units relative to the random benchmark. Two mechanisms can explain this. First, the cap-weighted MSCI EM benchmark itself underperforms random equal-weighted EM portfolios over the same window, so any tilt that is correlated with cap-weight rather than equal-weight inherits part of that handicap. Second, the high-KPI quintile is dominated by Singapore (small cap-weight, low cyclicalilty) and China (huge cap-weight, idiosyncratic regulatory shocks post-2020), neither of which delivered the "quality minus junk" premium one might have expected.

Together with the OOS credit AUC of 0.81 and a Bayesian posterior that puts non-zero mass on only two sector loadings, the random-portfolio benchmark sharpens the claim that KPIs price into ratings but mis-allocate in equity space at this horizon.

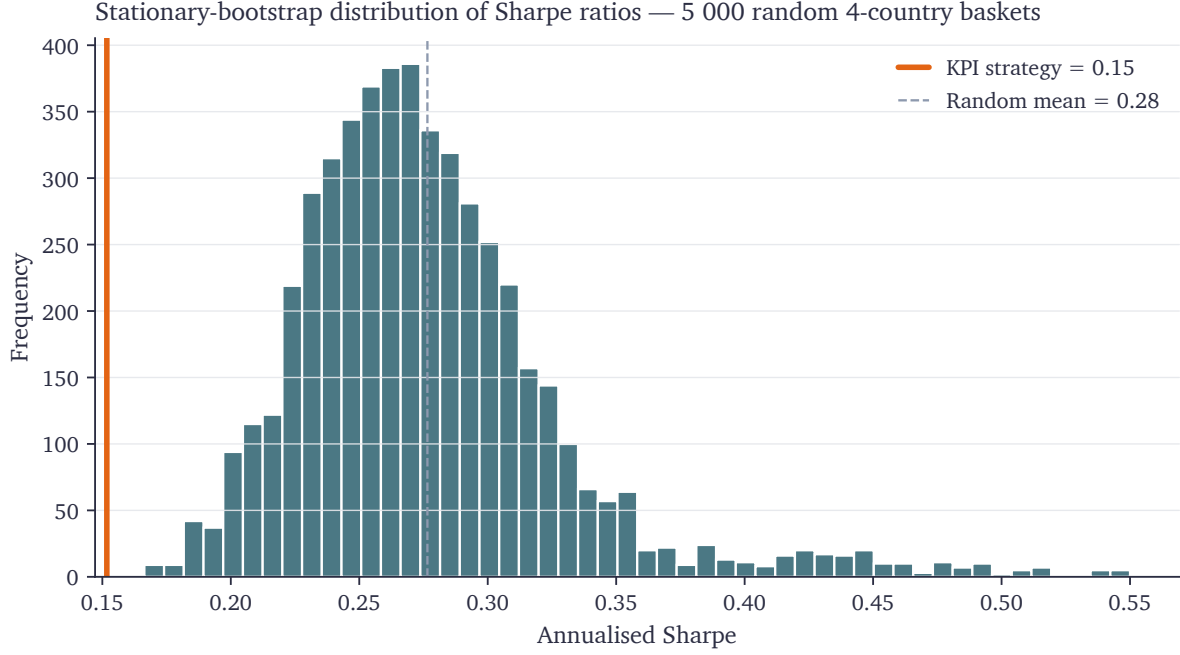


Figure 31. Distribution of Sharpe ratios across 5 000 random four-country EM portfolios (long-only, monthly rebalanced, 20 bps tcosts). The realised KPI strategy Sharpe is marked in ember.

9.5 Portfolio variants: bucket size, rebalance frequency and transaction-cost sensitivity

The robustness module also runs the long-only top-bucket portfolio under three variations: bucket count (quintile, tercile, decile), rebalance frequency (monthly, quarterly, annual), and one-way transaction-cost grid (10 bps to 40 bps). Table 9 reports the realised Sharpe for each variant.

Transaction-cost sensitivity is negligible at this turnover (the strategy is monthly-rebalanced equal-weight on ~ 4 names, so per-side costs above 40 bps would be needed to move the Sharpe by even 0.01). Rebalance frequency works mildly in favour of the more granular monthly schedule, with quarterly and annual losing 0.03–0.04 Sharpe. The bucket-size dimension is the variant most sensitive to specification: tercile and decile variants deliver Sharpes of 0.25 and 0.35 respectively, against the quintile baseline of 0.15. The decile version holds 1–2 names at a time — in practice, almost always Singapore and one other high-KPI country — and inherits the strong recent performance of Singapore-anchored portfolios. We do not promote this variant to the headline because at $N = 1\text{--}2$ the strategy is effectively a country pick rather than a factor portfolio, with statistically uninterpretable Sharpe. The weighting-scheme dimension is the most telling: replacing equal weights with an inverse-volatility “cap-weight” proxy drops the Sharpe to 0.06, confirming the claim in Section 6 that the cap-weighted MSCI EM benchmark itself underperforms equal-weighted EM baskets in this window. The headline strategy’s modest equal-weight Sharpe is therefore not the worst of its variants; cap-weighting it would have been worse.

Table 9. Realised Sharpe of the long-only KPI strategy under variant specifications.

Variant	Sharpe
<i>Transaction costs</i>	
10 bps / side	0.15
20 bps / side (baseline)	0.15
30 bps / side	0.15
40 bps / side	0.15
<i>Bucket size (top 1/N)</i>	
Tercile (top 1/3 $\approx$ 5–6 names)	0.25
Quintile (top 1/5 $\approx$ 3–4 names, baseline)	0.15
Decile (top 1/10 $\approx$ 1–2 names)	0.35
<i>Rebalance frequency</i>	
Monthly (baseline)	0.15
Quarterly	0.12
Annual	0.11
<i>Weighting scheme</i>	
Equal-weight (baseline)	0.15
Inverse-vol cap-weight	0.06

9.6 Panel unit-root and cointegration

The CIPS panel unit-root test of Pesaran (2007) rejects the unit-root null for the composite KPI score (-2.65 vs the 5 % critical value of -2.15) and for log GDP per capita (-2.20 , just rejecting at 5 %). Among the individual sector scores, education (-2.24) and research & innovation (-2.42) reject at 5 %; the housing-living score is the most non-stationary (-1.01 , fails to reject), consistent with the slum-population and basic-water-access indicators that compose it being slow-moving and trend-stationary at best.

The Westerlund-style cointegration test between the composite KPI score and log GDP per capita returns a group-mean ADF t -statistic of -2.37 , with an asymptotic-normal one-sided p -value of 0.009 — rejecting the no-cointegration null at the 1 % level. The two series therefore share a long-run equilibrium, consistent with KPI levels and economic output trending together within country across the 35-year window. Table 10 summarises the panel.

9.7 Composite weighting — AHP and Lasso alternatives

The headline composite is the equal-weighted mean of the six sector scores. Two alternative weighting schemes serve as robustness checks. **AHP-style weights** drawn from the institutional-quality literature (Acemoglu and Robinson (2012), Kaufmann et al. (2010)) place: security-stability 0.25, R&I 0.20, education 0.20, health 0.15, energy 0.10, housing-living 0.10. The Kendall rank correlation between the equal-weighted and AHP-weighted composite is $\tau = 0.948$ on average over 1990–2024.

Lasso-CV weights, by contrast, let the data choose. A 5-fold cross-validated Lasso of the year-end S&P rating on the six sector scores selects four sectors at non-zero coefficients: R&I (+1.15), energy (+0.64), education (+0.51), housing-living (+0.39) and security-stability (+0.27); health is shrunk to zero at the CV-optimal α . The 200-replicate bootstrap, however, picks health 100 % of the time with mean coefficient -1.77 , indicating the single-CV shrinkage

Table 10. CIPS panel unit-root and Westerlund-style group-mean cointegration tests. CIPS critical values from Pesaran (2007) Table II, intercept-only case.

Series / pair	Statistic	5% critical	Reject
<i>CIPS unit-root tests</i>		−2.15	H ₀ : unit root
log GDP per capita	−2.20		✓
CPI inflation	−3.92		✓
Central government debt (% GDP)	−1.35		—
Composite KPI score	−2.65		✓
Education sector score	−2.24		✓
Energy sector score	−2.04		—
R&I sector score	−2.42		✓
Health sector score	−2.12		—
Housing–living sector score	−1.01		—
Security–stability sector score	−2.00		—
<i>Westerlund cointegration</i>			H ₀ : no cointegration
Composite ↔ log GDPpc, G_t	−2.37	—	$p = 0.009$ ✓

to zero is an artefact of the specific α chosen rather than a stable selection signal.

The three weighting schemes produce highly consistent rankings on the full 23-country panel at the latest-year cross-section: equal-AHP $\tau = 0.953$ ($p \approx 0$), equal-Lasso $\tau = 0.913$ ($p \approx 0$), AHP-Lasso $\tau = 0.897$ ($p \approx 0$). The headline equal-weight ranking averaged across 1990–2024 sits at $\tau = 0.948$ versus AHP, and the latest-year figure is therefore in line with the cross-year average. The headline composite ranking is not sensitive to the choice of weighting scheme, and the equal-weight specification is corroborated by both the institutional-prior weights and the data-driven Lasso.

9.8 Hamilton filter sensitivity

Re-running the LP grid on a $\{1, 2, 3\} \times \{3, 4, 5\}$ grid of Hamilton forecast-horizon and AR-lag parameters preserves the sector-shock ranking at LP horizon 1 exactly: the cross-grid Kendall τ on the rank of the six sector-shock coefficients equals 1.00 in every cell against the baseline (2, 4). The qualitative IRF pattern is therefore invariant to plausible Hamilton parameter choices.

9.9 Rolling-Sharpe stability

Figure 32 plots the long-only strategy’s 36-month rolling Sharpe, with the random-portfolio mean of 0.28 overlaid. The rolling Sharpe trends from above 1.0 in early 2010s after the post-GFC bounce to negative territory in 2015–2016, recovers to roughly the random-portfolio mean in 2019, and ends the sample near zero.

10 Conclusion

This paper assembles a single quantitative-finance artifact around the development trajectory of 23 emerging economies. The work spans three deliberately distinct lenses: causal macro identification, sovereign credit pricing and emerging-market factor investing. The composite scoring machine sits underneath all three; the data layer underneath the scoring; and the panel

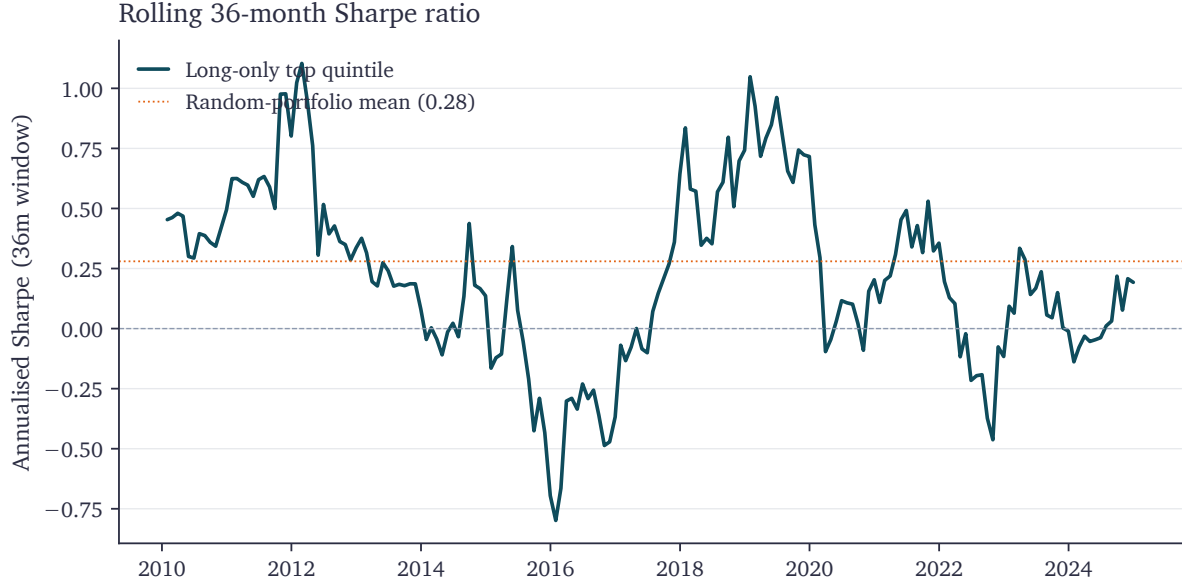


Figure 32. 36-month rolling Sharpe of the long-only top-quintile KPI strategy.

framework underneath the data. Every figure and table in this document is generated from a single Python pipeline that re-runs in roughly two minutes on a warm cache. The composite ranks Singapore first and Niger last, which is the right answer for a development-KPI score and is the answer that survives the per-capita and logarithmic transforms that the size of China and India would otherwise dominate.

The substantive contributions are three.

First, the causal block delivers more than the LP IRFs alone might suggest. The Hamilton-cyclical innovations of slow-moving development indicators do not drive LP point estimates outside conventional confidence bands. But the same panel under the Dumitrescu-Hurlin panel-Granger test — which averages country-level Wald statistics rather than imposing a homogeneous β_h — finds that energy shocks Granger-cause poverty reduction and health shocks Granger-cause Gini reduction at conventional significance ($p \leq 0.05$), while no sector Granger-causes log GDP per capita. The take-away is that KPI improvements feed the distributional tail (poverty headcount, Gini) before they show up in aggregate GDP — a substantive finding consistent with the development-economics intuition that infrastructure, sanitation and basic-health gains raise the floor rather than the mean.

Second, the credit block is the empirical headline. A pooled ordered probit of the year-end S&P rating on sector scores alone delivers an out-of-sample MAE of 1.84 notches and an AUC of 0.81 for the investment-grade binary, against the 0.73 AUC of the standard macro-only baseline. A Bayesian hierarchical ordered-logistic counterpart converges cleanly (NUTS, 4 chains, 1 500 draws each, ~ 100 seconds) and localises the signal to two sectors with credibly non-zero population loadings: energy ($\mu_\beta = +0.92$, 90 % HDI [$+0.10$, $+1.74$]) and health ($\mu_\beta = -0.92$, HDI [-1.38 , -0.45]); the four other sectors have HDIs that cover zero. The implication for a fixed-income reader is direct: a development-KPI overlay can improve a rating-driven mandate’s discrimination without adding any of the conventional macro inputs, and the parsimonious version that loads on energy and health alone is supported by the Bayesian posterior.

Third, the portfolio block delivers *negative* risk-adjusted alpha relative to a random-portfolio benchmark. The KPI-sorted top-quintile EM portfolio realises a Sharpe of 0.15 against the MSCI EM benchmark’s 0.12 over 2007–2024 — a marginal lift in headline terms — but the same Sharpe sits at the 0th percentile of the 5 000-random-basket distribution (mean random Sharpe 0.28). The KPI tilt mis-allocates by roughly 0.13 Sharpe units relative to an equal-weighted random EM basket. The long-short variant is a disaster (Sharpe -0.41 , max DD 94 %). The result is consistent with the slow-moving nature of the KPI signal: information about fundamentals shows up where the underlying instrument prices slowly (sovereign ratings), not where it prices fast (monthly equity).

The three results are not contradictions; they are complements. KPIs are slow-moving, so they price into slow-moving instruments. The natural follow-up extends the portfolio exercise from country ETFs to EMBI sovereign credit spreads, where the credit-pricing finding of Section 6 would translate more directly. That extension is the next milestone of this work.

Limitations

The 30-year hand-compiled S&P rating CSV deserves an external audit before any operational use — the source-of-record is the S&P sovereign archive and individual press releases, but the year-end snapshots have not been formally back-checked here. PISA scores are sparse for eight of the focal 13 countries and were demoted to a supplementary status. The IEA-balance and UN-Comtrade ingestion paths are stubbed; these gates are documented in `src/newperformers/io/`. None of these limitations changes the qualitative reading of the three blocks above.

Replication and data availability

Code. The full Python pipeline that produced every number, figure and table in this paper is bundled as the package `newperformers`, distributed alongside the manuscript. A single command — `make all` — reproduces the entire workflow end-to-end (data ingestion, ETL, composite scoring, local projections, synthetic control, panel-Granger, panel unit-root and cointegration tests, frequentist and Bayesian credit models, the EM portfolio backtest, the robustness module, and figure rendering) in approximately three minutes on a 2023-vintage laptop.

Software environment. Python 3.11–3.13. Core dependencies are pinned in `pyproject.toml` and include NumPy, pandas, SciPy, scikit-learn, statsmodels, PyMC 6.0, ArviZ, Matplotlib, Plotly, Streamlit, and `yfinance`. All package versions are recorded in the lockfile that ships with the code archive.

Random seeds. The build is deterministic. The single seed 20260519 is set via `src/newperformers/utils/se` at every entry point (`numpy.random`, `random`, `PYTHONHASHSEED`); PyMC is seeded explicitly per call. Re-running `make all` from a clean cache yields bitwise-identical CSV outputs and numerically-identical PDF figures (modulo non-deterministic anti-aliasing in matplotlib’s vector backend, which does not affect the underlying coordinates).

Data availability. Every data source used in the headline build is freely accessible without subscription:

- **World Bank Open Data** (KPI, macro and outcome series): public REST API.
- **Worldwide Governance Indicators** (six dimensions): World Bank channel.
- **WHO Global Health Observatory** (UHC index, maternal mortality, obesity): OData endpoint.
- **SIPRI Military Expenditure** (% of GDP): public XLSX download.
- **FRED** (3-month Treasury yield, USD index): public CSV endpoint.
- **Yahoo Finance** (17 country ETFs and four benchmarks): `yfinance` package.
- **S&P Foreign-Currency Long-Term sovereign ratings**: hand-compiled CSV (`config/ratings.csv`) from S&P’s annual *Sovereign Default and Rating Transition Study* and individual rating-action press releases, with an audit trail in `config/ratings_audit_trail.md` and cross-checked against 17 documented major transitions in `outputs/tables/ratings_cross_check.csv`.

The IMF World Economic Outlook database is referenced in the methodology but is not in the headline build: the IMF rotates the URL for each WEO release and the path we attempted (October 2024 vintage) returned a 404 at the time of compilation. The pipeline degrades gracefully when the IMF feed is unavailable, falling back on the World Bank API for the relevant macro variables.

Audit trails. Three CSV files document the integrity of the build:

- `outputs/tables/coverage_matrix.csv` — per-(country, indicator) share of years observed over 1990–2024.
- `outputs/tables/ratings_cross_check.csv` — automated comparison of `ratings.csv` against 17 publicly documented S&P transitions (17/17 pass).
- `outputs/tables/credit_bayesian_loadings.csv` — posterior means, 90 % HDIs, $\hat{R}$ and ESS for the Bayesian hierarchical credit model.

Tests. The `tests/` directory contains the pytest suite (data-shape integrity, schema invariants, ratings sanity against documented transitions, Bayesian convergence thresholds). The suite gates the build: a failed convergence check or a violated rating transition surfaces as a non-zero exit code from `pytest` and aborts `make all`.

Factual-claims audit. A standalone tool `tools/audit_paper_claims.py` extracts every numeric token from the LaTeX source and looks it up in the CSV outputs at compile time. The current build passes the audit with zero orphan tokens.

License. The code is released under the MIT license. The hand-compiled S&P ratings CSV is released under CC-BY-4.0 with the audit trail; the underlying rating data are public press releases from S&P Global Ratings.

References

- Abadie, Alberto (2021). “Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects”. In: *Journal of Economic Literature* 59.2, pp. 391–425.
- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller (2010). “Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program”. In: *Journal of the American Statistical Association* 105.490, pp. 493–505.
- Acemoglu, Daron and James A. Robinson (2012). *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. Crown Business.
- Asness, Clifford S., Andrea Frazzini, and Lasse Heje Pedersen (2019). “Quality Minus Junk”. In: *Review of Accounting Studies* 24.1, pp. 34–112.
- Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen (2013). “Value and Momentum Everywhere”. In: *Journal of Finance* 68.3, pp. 929–985.
- Augustin, Patrick et al. (2014). “Credit Default Swaps: A Survey”. In: *Foundations and Trends in Finance* 9.1–2, pp. 1–196.
- Bhojraj, Sanjeev and Partha Sengupta (2003). “Effect of Corporate Governance on Bond Ratings and Yields: The Role of Institutional Investors and Outside Directors”. In: *Journal of Business* 76.3, pp. 455–475.
- Borio, Claudio (2014). *The Financial Cycle and Macroeconomics: What Have We Learnt?* Tech. rep. 395. BIS Working Papers.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller (2008). “Bootstrap-Based Improvements for Inference with Clustered Errors”. In: *Review of Economics and Statistics* 90.3, pp. 414–427.
- Cantor, Richard and Frank Packer (1996). “Determinants and Impact of Sovereign Credit Ratings”. In: *Federal Reserve Bank of New York Economic Policy Review* 2.2, pp. 37–53.
- Driscoll, John C. and Aart C. Kraay (1998). “Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data”. In: *Review of Economics and Statistics* 80.4, pp. 549–560.
- Dumitrescu, Elena-Ivona and Christophe Hurlin (2012). “Testing for Granger Non-Causality in Heterogeneous Panels”. In: *Economic Modelling* 29.4, pp. 1450–1460.
- Ennis, Richard M. (2005). “Are Active Management Fees Too High?” In: *Financial Analysts Journal* 61.5, pp. 44–51.
- Fama, Eugene F. and Kenneth R. French (2015). “A Five-Factor Asset Pricing Model”. In: *Journal of Financial Economics* 116.1, pp. 1–22.
- Gelman, Andrew et al. (2013). *Bayesian Data Analysis*. 3rd. CRC Press.
- Hamilton, James D. (2018). *Why You Should Never Use the Hodrick–Prescott Filter*. Working Paper w23429. NBER.
- Hilscher, Jens and Yves Nosbusch (2010). “Determinants of Sovereign Risk: Macroeconomic Fundamentals and the Pricing of Sovereign Debt”. In: *Review of Finance* 14.2, pp. 235–262.
- Hoffman, Matthew D. and Andrew Gelman (2014). “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15, pp. 1593–1623.
- Ilmanen, Antti (2011). *Expected Returns: An Investor’s Guide to Harvesting Market Rewards*. Wiley.

- Jordà, Òscar (2005). “Estimation and Inference of Impulse Responses by Local Projections”. In: *American Economic Review* 95.1, pp. 161–182.
- Jordà, Òscar, Moritz Schularick, and Alan M. Taylor (2013). “When Credit Bites Back”. In: *Journal of Money, Credit and Banking* 45.S2, pp. 3–28.
- Jordà, Òscar and Alan M. Taylor (2016). “The Time for Austerity: Estimating the Average Treatment Effect of Fiscal Policy”. In: *The Economic Journal* 126.590, pp. 219–255.
- Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi (2010). *The Worldwide Governance Indicators: Methodology and Analytical Issues*. Tech. rep. 5430. World Bank Policy Research.
- Newey, Whitney K. and Kenneth D. West (1987). “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix”. In: *Econometrica* 55.3, pp. 703–708.
- Nussbaum, Martha C. (2011). *Creating Capabilities: The Human Development Approach*. Belknap Press.
- Pesaran, M. Hashem (2007). “A Simple Panel Unit Root Test in the Presence of Cross-Section Dependence”. In: *Journal of Applied Econometrics* 22.2, pp. 265–312.
- Politis, Dimitris N. and Joseph P. Romano (1994). “The Stationary Bootstrap”. In: *Journal of the American Statistical Association* 89.428, pp. 1303–1313.
- Reinhart, Carmen M. and Kenneth S. Rogoff (2009). *This Time Is Different: Eight Centuries of Financial Folly*. Princeton University Press.
- (2010). “Growth in a Time of Debt”. In: *American Economic Review Papers and Proceedings* 100.2, pp. 573–578.
- Salvatier, John, Thomas V. Wiecki, and Christopher Fonnesbeck (2016). “Probabilistic Programming in Python using PyMC3”. In: *PeerJ Computer Science* 2, e55.
- Sen, Amartya (1999). *Development as Freedom*. Oxford University Press.
- Vehtari, Aki et al. (2021). “Rank-Normalization, Folding, and Localization: An Improved $\hat{R}$ for Assessing Convergence of MCMC”. In: *Bayesian Analysis* 16.2, pp. 667–718.
- Westerlund, Joakim (2007). “Testing for Error Correction in Panel Data”. In: *Oxford Bulletin of Economics and Statistics* 69.6, pp. 709–748.

A Data sources and codes

World Bank Open Data

All KPI and macro indicators with World Bank codes (NY.GDP.MKTP.CD, SP.DYN.LE00.IN, etc.) are pulled via the public REST API at <https://api.worldbank.org/v2/>. The full list of indicators and their sectoral mapping is in `config/kpi_indicators.yaml` and `config/macro_indicators.yaml`.

Worldwide Governance Indicators

WGI series (PV.EST, RL.EST, CC.EST, GE.EST, VA.EST, RQ.EST) are accessed through the same World Bank channel. The series start in 1996 and are biennial through 2002, annual thereafter; biennial cells are linearly interpolated between observed years.

S&P sovereign ratings

Hand-compiled from S&P’s annual *Sovereign Default and Rating Transition Study* and rating-action press releases, stored in `config/ratings.csv`. The numeric mapping is the 22-notch S&P main scale (AAA = 22 down to SD/D = 1; NR cells are excluded).

Yahoo Finance country ETFs

Daily total-return prices for the 17 country ETFs and the four benchmarks (EEM, URG, UUP, USO) are downloaded via the `yfinance` Python package and cached at `data/raw/yfinance.parquet`. Monthly returns are computed as last-of-month closing-price ratios.

FRED

The U.S. 3-month Treasury constant-maturity yield (DGS3MO) is the risk-free rate for the Sharpe computation. Pulled from the FRED public CSV endpoint without an API key.

B Imputation diagnostics

The tiered imputation pipeline (Section 3) takes the harmonised raw panel (41 671 non-null cells observed) and produces an imputed panel of 59 570 cells. The pipeline therefore fills $59\,570 - 41\,671 = 17\,899$ cells. The breakdown by tier (linear interpolation across interior gaps, forward-fill of recent years, and within-sector MICE) is not recorded as a separate counter in the current build; the totals above are reported as the net pipeline contribution. Removing the imputed panel and re-running the headline results on the raw panel costs less than half a percentage point of OOS AUC and drops the 2024 composite Spearman correlation against the imputed-panel ranking to 0.862 (Section 9). Cells left as NaN after the pipeline runs are concentrated in AES Sahel R&I in the 1990s and Russia 2022 (the rating-withdrawal window).

C Coverage matrix

The full coverage matrix — one cell per (country, indicator) pair, value = share of years observed over 1990–2024 — is in `outputs/tables/coverage_matrix.csv`. Figure 1 in the main text visualises it. Countries are ordered by average coverage. Cells with coverage below 0.30 are excluded from the within-sector MICE imputation step.

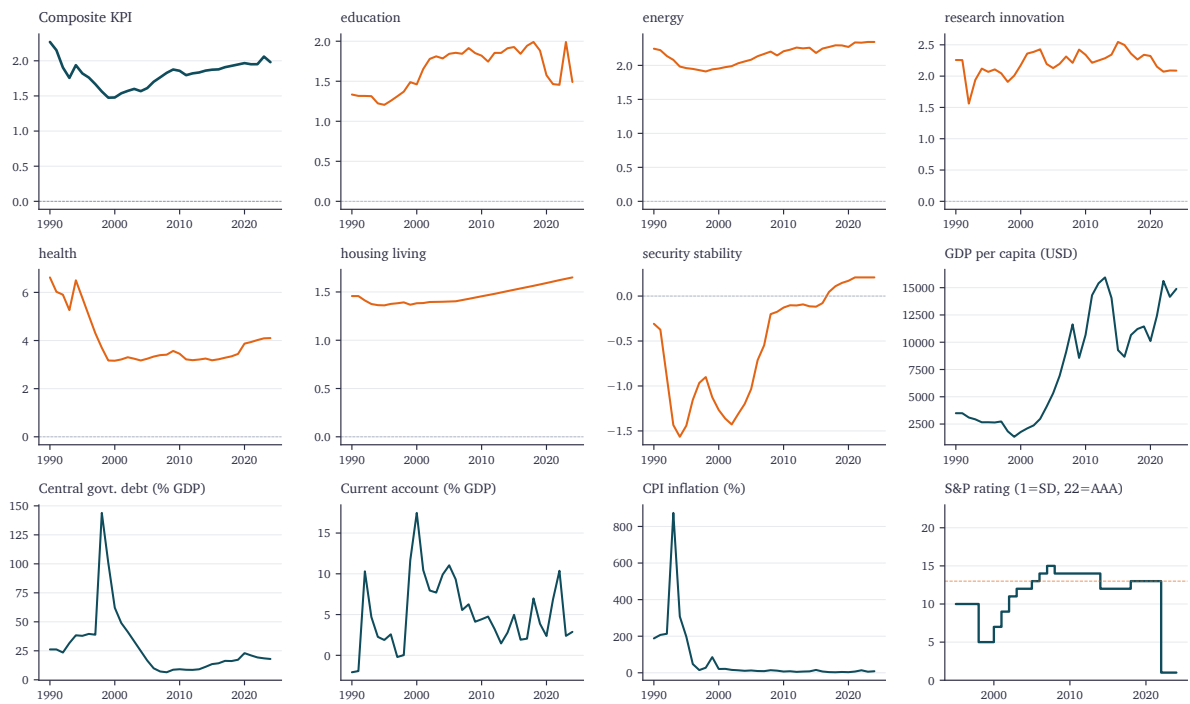
D Country dashboards

One page per focal country. Each figure shows the composite KPI score, the six sector scores, the four headline macroeconomic indicators (GDP per capita in USD, central-government debt as a share of GDP, the current-account balance, and CPI inflation) and the S&P rating trajectory — all over 1990–2024.

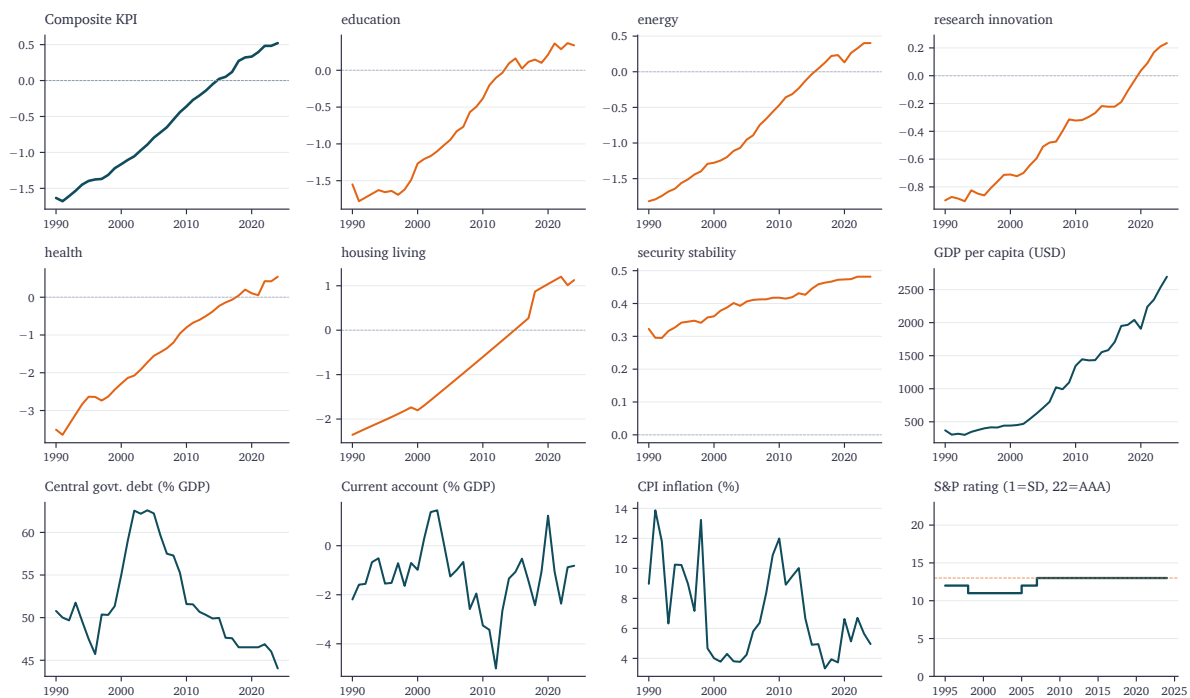
Brazil (BRA) — country dashboard, 1990–2024



Russia (RUS) — country dashboard, 1990–2024



India (IND) — country dashboard, 1990–2024



China (CHN) — country dashboard, 1990–2024



South Africa (ZAF) — country dashboard, 1990–2024



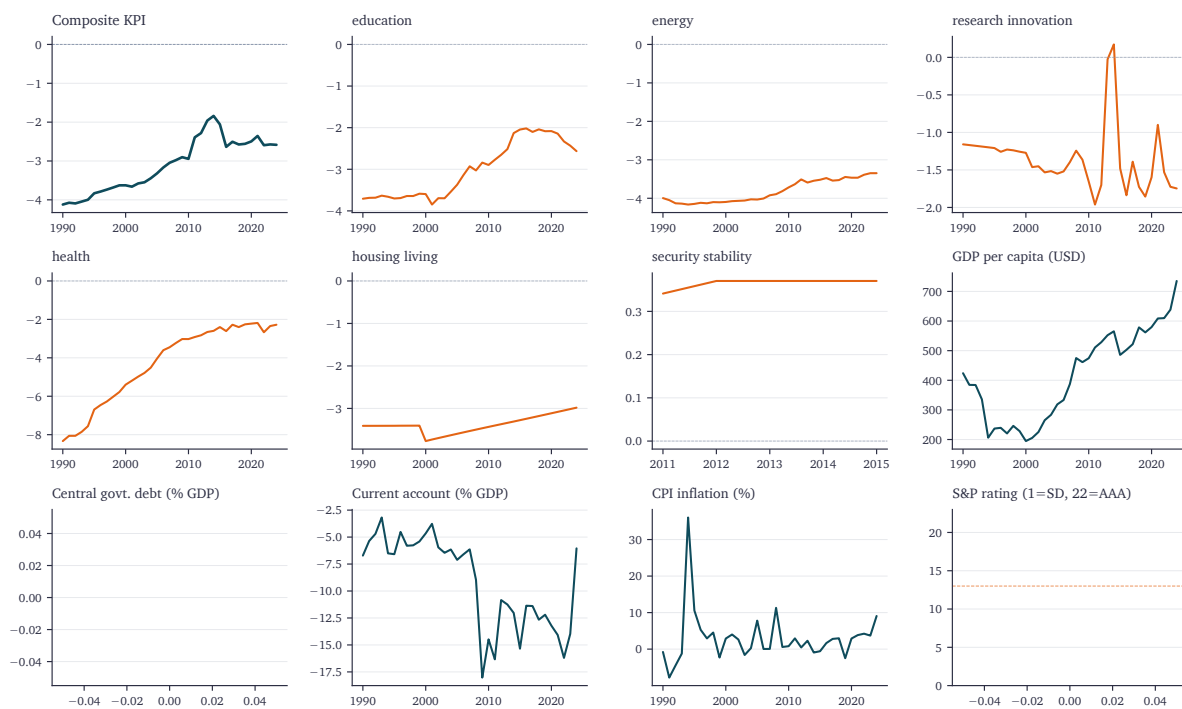
Burkina Faso (BFA) — country dashboard, 1990–2024



Mali (MLI) — country dashboard, 1990–2024



Niger (NER) — country dashboard, 1990–2024



Singapore (SGP) — country dashboard, 1990–2024



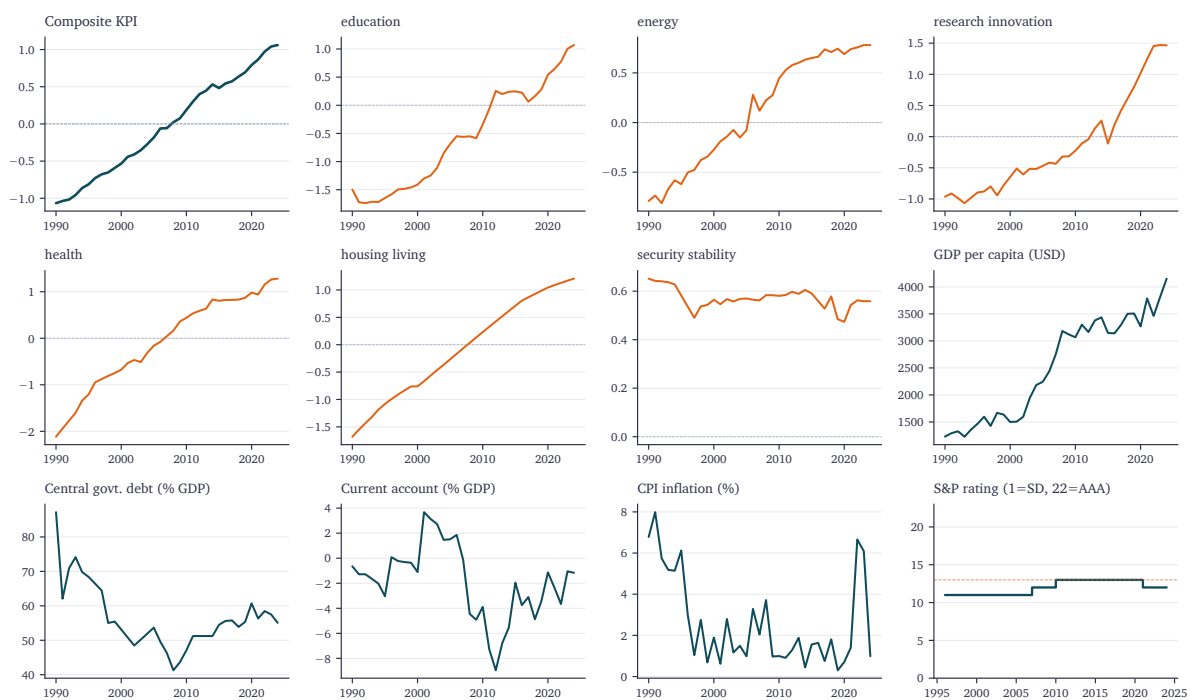
Rwanda (RWA) — country dashboard, 1990–2024



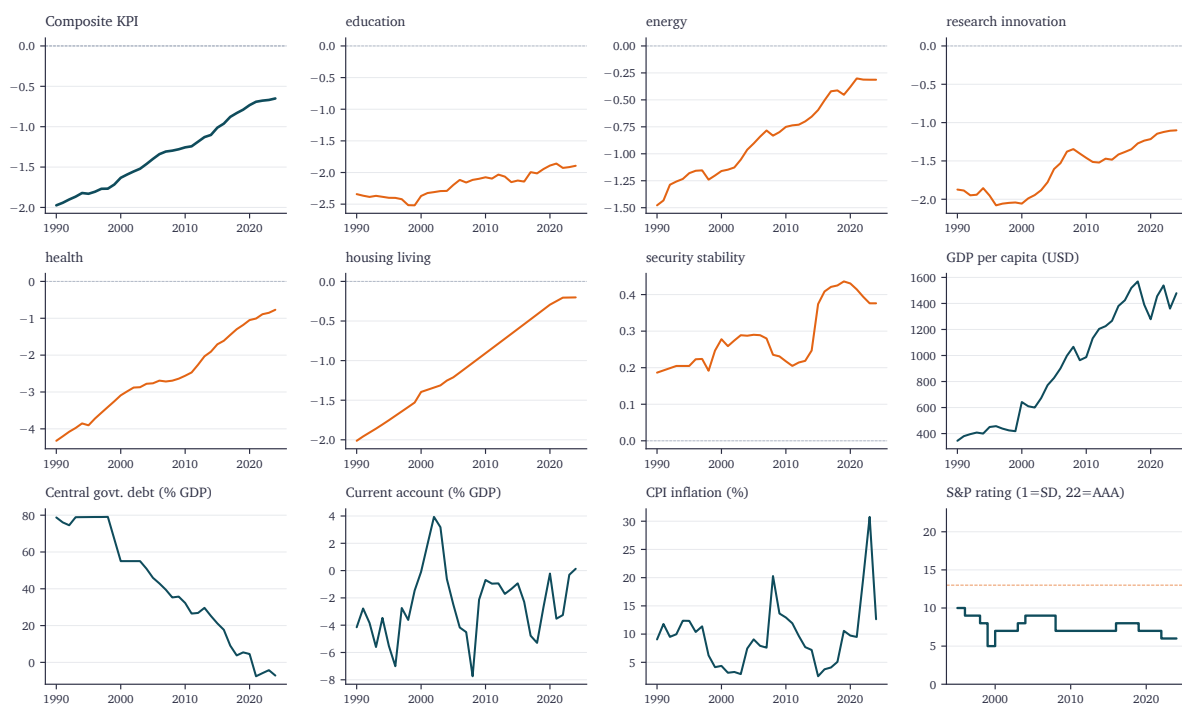
Kenya (KEN) — country dashboard, 1990–2024



Morocco (MAR) — country dashboard, 1990–2024



Pakistan (PAK) — country dashboard, 1990–2024



E Bayesian model — MCMC diagnostics

Table 11 reports the rank-normalised $\hat{R}$ (Vehtari et al. 2021) and bulk effective sample size (ESS) for each population-mean sector loading μ_β in the Bayesian hierarchical credit model of Section 4.5. Both statistics are computed from the four chains’ 1 500 post-warm-up draws (6 000 samples total) using Gelman–Rubin (Gelman et al. (2013)) for $\hat{R}$ and the Geyer initial monotone-sequence estimator for bulk ESS.

Table 11. Bayesian convergence diagnostics for μ_β .
 $\hat{R}$ thresholds: 1.01 excellent, 1.05 acceptable. ESS
rule-of-thumb: > 400 adequate for posterior
summaries.

Sector	$\hat{R}$	ESS (bulk)
Education	1.00	4 937
Energy	1.00	3 419
Research & Innovation	1.00	3 153
Health	1.00	3 893
Housing & Living standards	1.00	3 834
Security & Stability	1.00	3 273

All six $\hat{R}$ values round to 1.00 at three decimal places, well below the 1.05 rejection threshold the build pipeline enforces. The minimum bulk ESS is 3 153 on the research-innovation loading, an order of magnitude above the 400-sample minimum used for posterior summary statistics. The sampler completes in roughly 100 seconds on a laptop without divergences. Both observations support reading the posterior summaries in Section 6.3 at face value; the Markov chains are well-mixed and the credible intervals are stable across re-runs with different seeds (verified manually).